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Chapter 1

Introduction

In many natural phenomena and complex evolving systems, it is observed that on ‘average’ the systems remain the same. Martingale theory, in discrete and continuous time, is of tremendous importance in the analysis of such systems.

In this project, we study martingales in discrete time. In Chapter 2, we discuss the concept of Conditional Expectation and in Chapter 3, we set up the basic definitions used throughout this report.

In Chapter 4, we study the important convergence theorems related to Martingale theory. This leads us to the study of $\mathcal{L}^2$ martingales in Chapter 5. In Chapters 6 and 7, we explore the relation between $\mathcal{L}^1$ convergence and Uniform integrability. We finish the report with an application to discrete Black-Scholes model, which analyzes option pricing.

Chapter 2

Conditional Expectation

2.1 Introduction

In this chapter, we discuss the concept of conditional expectation. The main reference for this chapter is [1, Chapter 8] and [2, Chapter 9].

2.2 Motivation

Let X and Z be two discrete random variables defined on $(\Omega, \mathcal{F}, \mathbb{P})$. X takes the values $x_1, x_2, \dots, x_m$ and Z takes $z_1, z_2, \dots, z_n$. Partition Ω into sets such that Z is constant in each particular set. Therefore, the σ -algebra $\mathcal{G} = \sigma(Z)$ consists of sets $\{Z \in B\}$, $B \in \mathcal{B}$.

Now, if $Z(\omega) = z_j$, then $Y(\omega) := \mathbb{E}(X|Z = z_j) =: y_j$ (say)

This implies that Y is constant on Z -atoms, which in turn means that Y is $\mathcal{G}$ -measurable.

$$\begin{aligned} \int_{\{Z=z_j\}} Y d\mathbb{P} &= y_j \mathbb{P}(Z = z_j) = \sum_i x_i \mathbb{P}(X = x_i | Z = z_j) \mathbb{P}(Z = z_j) \\ &= \sum_i x_i \mathbb{P}(X = x_i, Z = z_j) = \int_{\{Z=z_j\}} X d\mathbb{P} \end{aligned}$$

Let $G_j = \{Z = z_j\}$ then, $\mathbb{E}(Y I_{G_j}) = \mathbb{E}(X I_{G_j})$.

Thus $\forall G \in \mathcal{G}$, I_G is the sum of I_{G_j} 's.

We get $\mathbb{E}(Y I_G) = \mathbb{E}(X I_G)$ which in turn implies, $\int_G Y d\mathbb{P} = \int_G X d\mathbb{P} \quad \forall G \in \mathcal{G}$.

Theorem 2.2.1 (Fundamental Theorem and definition (Kolmogorov, 1933)[2, 9.2]). *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and X be a r.v. with $\mathbb{E}(|X|) < \infty$. Let $\mathcal{G}$ be a sub σ -algebra of $\mathcal{F}$. Then there exists a r.v. Y such that*

- a) Y is $\mathcal{G}$ -measurable.
- b) $\mathbb{E}(|Y|) < \infty$
- c) $\int_G Y d\mathbb{P} = \int_G X d\mathbb{P} \quad \forall G \in \mathcal{G}$.

Moreover, if $\tilde{Y}$ is another r.v. with these properties then, $\tilde{Y} = Y$ a.s. where $Y = \mathbb{E}(X|\mathcal{G})$ a.s.

Proof. We firstly show the almost sure uniqueness of $\mathbb{E}(X|\mathcal{G})$. Suppose $X \in \mathcal{L}^1$ and that Y and $\tilde{Y}$ are versions of $\mathbb{E}(X|\mathcal{G})$. Then $Y, \tilde{Y} \in \mathcal{L}^1(\Omega, \mathcal{G}, \mathbb{P})$ and $\mathbb{E}(Y - \tilde{Y}; G) = 0 \quad \forall G \in \mathcal{G}$.

Suppose Y and $\tilde{Y}$ are not a.s. equal. We may assume $\mathbb{P}(Y > \tilde{Y}) > 0$.

Since $\{Y > \tilde{Y} + \frac{1}{n}\} \uparrow \{Y > \tilde{Y}\}$, $\mathbb{P}(Y > \tilde{Y} + \frac{1}{n}) > 0$ for some n .

But $\{Y - \tilde{Y} > \frac{1}{n}\} \in \mathcal{G}$ as both Y and $\tilde{Y}$ are $\mathcal{G}$ -measurable.

$\mathbb{E}(Y - \tilde{Y}; Y - \tilde{Y} > \frac{1}{n}) \geq \frac{1}{n} \mathbb{P}(Y - \tilde{Y} > \frac{1}{n}) > 0$, which is a contradiction. Therefore, $Y = \tilde{Y}$ a.s.

Now , the existence of $\mathbb{E}(X|\mathcal{G})$ for $X \in \mathcal{L}^2$ is shown. Suppose $X \in \mathcal{L}^2 := \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$.

Let $\mathcal{G}$ be a sub-algebra of $\mathcal{F}$ and let $\mathcal{K} := \mathcal{L}^2(\Omega, \mathcal{G}, \mathbb{P})$. Now $\mathcal{K}$ is complete in $\mathcal{L}^2$ norm. We know, there exists Y in $\mathcal{K} := \mathcal{L}^2(\mathcal{G})$ such that

$$a) \mathbb{E}(X - Y)^2 = \inf[\mathbb{E}(X - W)^2; W \in \mathcal{L}^2(\mathcal{G})]$$

$$b) \langle X - Y, Z \rangle = 0 \quad \forall \quad Z \in \mathcal{L}^2(\mathcal{G})$$

If $G \in \mathcal{G}$, then $Z := I_G \in \mathcal{L}^2(\mathcal{G})$ and (b) states that $\mathbb{E}(Y; G) = \mathbb{E}(X; G)$. Hence Y is a version of $\mathbb{E}(X|\mathcal{G})$.

Lastly, we show the existence of $\mathbb{E}(X|\mathcal{G})$ for $X \in \mathcal{L}^1$. Now $X = X^+ - X^-$. It is enough to deal with $X \in (\mathcal{L}^1)^+$. Let $X \in (\mathcal{L}^1)^+$, we can choose a bounded variable X_n with $0 \leq X_n \uparrow X$. Since $X_n \in \mathcal{L}^2$, we can choose a version of Y_n of $\mathbb{E}(X_n|\mathcal{G})$.

Need to show $0 \leq Y_n \uparrow a.s.$ □

Lemma 2.2.2. *If U is a non-negative bounded r.v. , then $\mathbb{E}(U|\mathcal{G}) \geq 0 a.s.$*

Proof. Let W be a version of $\mathbb{E}(U|\mathcal{G})$. If $\mathbb{P}(W < 0) > 0$, then for some n the set $G = \{W < -\frac{1}{n}\}$ has positive probability i.e. $\mathbb{P}(W < -\frac{1}{n}) > 0$ for some n , so that,

$$0 \leq \mathbb{E}(U; G) = \mathbb{E}(W; G) < -\frac{1}{n}\mathbb{P}(G) < 0 \quad \text{which is a contradiction}$$

Thus using this lemma 2.2.2 , we know $Y_n \geq 0$

Let $Y(\omega) := \limsup Y_n(\omega)$. Then Y is $\mathcal{G}$ -measurable and $Y_n \uparrow Y a.s.$

Using Monotone Convergence Theorem (MCT) , $\mathbb{E}(Y; G) = \mathbb{E}(X; G) \quad \forall \quad G \in \mathcal{G}$. □

Proposition 2.2.3 (Properties of Conditional Expectation [2, 9.7]). *Let a r.v. X satisfy $\mathbb{E}(|X|) < \infty$ and let $\mathcal{G}$ and $\mathcal{H}$ denote sub-algebras of $\mathcal{F}$.*

a) *If Y is any version of $\mathbb{E}(X|\mathcal{G})$ then $\mathbb{E}(Y) = \mathbb{E}(X)$.*

b) *If X is $\mathcal{G}$ - measurable , then $\mathbb{E}(X|\mathcal{G}) = X \quad a.s..$*

c) *$\mathbb{E}(a_1X_1 + a_2X_2|\mathcal{G}) = a_1\mathbb{E}(X_1|\mathcal{G}) + a_2\mathbb{E}(X_2|\mathcal{G}) a.s$*

d) *If $X \geq 0$, then $\mathbb{E}(X|\mathcal{G}) \geq 0 a.s.$*

e) *(Monotone) If $0 \leq X_n \uparrow X$, then $\mathbb{E}(X_n|\mathcal{G}) \uparrow \mathbb{E}(X|\mathcal{G}) a.s.$*

f) *(Conditional Fatou) If $X_n \geq 0$, then $\mathbb{E}(\liminf X_n|\mathcal{G}) \leq \liminf \mathbb{E}(X_n|\mathcal{G}) a.s.$*

g) *(Conditional DCT) If $|X_n(\omega)| \leq V(\omega) \forall n$ $\mathbb{E}(V) < \infty$ and $X_n \rightarrow X a.s.$, then $\mathbb{E}(X_n|\mathcal{G}) \rightarrow \mathbb{E}(X|\mathcal{G}) a.s.$*

h) *(Conditional Jensen) If $C : \mathbb{R} \rightarrow \mathbb{R}$ is convex and $\mathbb{E}(C|X|) < \infty$, then $\mathbb{E}(C(X)|\mathcal{G}) \geq C(\mathbb{E}(X|\mathcal{G})) a.s.$*

It is to be noted that $\mathbb{E}(X|\mathcal{G})\|_p \leq \|X\|_p$ for $p \geq 1$.

i) *(Tower) If $\mathcal{H}$ is a sub σ -algebra of $\mathcal{G}$, then $\mathbb{E}(\mathbb{E}(X|\mathcal{G})|\mathcal{H}) = \mathbb{E}(X|\mathcal{H}) a.s.$*

j) *If Z is $\mathcal{G}$ -measurable and bounded , then $\mathbb{E}(ZX|\mathcal{G}) = Z\mathbb{E}(X|\mathcal{G}) a.s..$ In particular, (b) follows.*

k) *(Role of independence) If $\mathcal{H}$ is independent of $\sigma(X, \mathcal{G})$, then $\mathbb{E}(X|\sigma(\mathcal{G}, \mathcal{H})) = \mathbb{E}(X|\mathcal{G}) a.s. .$ In particular if X is independent of $\mathcal{H}$, then $\mathbb{E}(X|\mathcal{H}) = \mathbb{E}(X) a.s.$*

Proof. a) $\mathbb{E}(Y; \Omega) = \mathbb{E}(X; \Omega)$ as $\Omega \in \mathcal{G}$ which implies that $\mathbb{E}(Y) = \mathbb{E}(X)$.

- b) Done in (j).
- c) Using the linearity of expectation , we can similarly prove the same for conditional expectation.
- d) Done in the Lemma 2.2.2.
- e) If $0 \leq X_n \uparrow X$, by lemma 2.2.2 $\forall n$, if Y_n is an version of $\mathbb{E}(X_n|\mathcal{G})$, then $0 \leq Y_n \uparrow a.s.$
Define $Y(\omega) := \limsup Y_n(\omega)$. Then Y is a $\mathcal{G}$ -measurable function and $Y_n \uparrow Y a.s.$
Now , $\mathbb{E}(Y_n; G) = \mathbb{E}(X_n; G) \quad \forall G \in \mathcal{G}$. By MCT, $\mathbb{E}(Y; G) = \mathbb{E}(X; G) \quad \forall G \in \mathcal{G}$.
- f) Let Y_n is any version of $\mathbb{E}(X_n|\mathcal{G})$. Define $X'_n := \inf_{k \geq n} X_k$ i.e. $X'_n \uparrow X' := \liminf X_n$.
By MCT, $\mathbb{E}(X'_n|\mathcal{G}) \uparrow \mathbb{E}(X'|\mathcal{G})$.
But $X'_n \leq X_n$. So,

$$\mathbb{E}(X'|\mathcal{G}) = \lim_{n \rightarrow \infty} \mathbb{E}(X'_n|\mathcal{G}) = \liminf \mathbb{E}(X_n|\mathcal{G}) \leq \liminf \mathbb{E}(X_n|\mathcal{G})$$

and hence $\mathbb{E}(\liminf X_n|\mathcal{G}) \leq \liminf \mathbb{E}(X_n|\mathcal{G})$ a.s..

- g) $X_n \rightarrow X$ implies that $X = \liminf X_n = \limsup X_n$. By Fatou's ,

$$\mathbb{E}(X|\mathcal{G}) = \mathbb{E}(\liminf X_n|\mathcal{G}) \leq \liminf \mathbb{E}(X_n|\mathcal{G}) \leq \limsup \mathbb{E}(X_n|\mathcal{G}) \leq \mathbb{E}(\limsup X_n|\mathcal{G}) = \mathbb{E}(X_n|\mathcal{G})$$

Thus $\liminf \mathbb{E}(X_n|\mathcal{G}) = \limsup \mathbb{E}(X_n|\mathcal{G})$ and then, $\mathbb{E}(X_n|\mathcal{G}) \rightarrow \mathbb{E}(X|\mathcal{G})$.

Alternatively, upon defining $Z_n = \sup_{k \geq n} |X_k - X|$ we get $Z_n \downarrow 0$.

Now, as $\mathbb{E}(X_n|\mathcal{G})$ and $\mathbb{E}(X|\mathcal{G})$ are finite as $\mathbb{E}(X_n) < \infty$ and

$$|\mathbb{E}(X_n|\mathcal{G}) - \mathbb{E}(X|\mathcal{G})| \leq \mathbb{E}(|X_n - X||\mathcal{G}) \leq \mathbb{E}(Z_n|\mathcal{G})$$

Thus we are done.

- h) There exists a countable sequence $\{(a_n, b_n)\}$ of points in $\mathbb{R}^2$ s.t. $c(x) = \sup_n (a_n x + b_n)$.
For fixed n , $c(x) \geq (a_n x + b_n)$, thus by lemma 2.2.2,

$$\mathbb{E}((c(X)|\mathcal{G}) \geq a_n \mathbb{E}(X|\mathcal{G}) + b_n. \quad (2.2.1)$$

As $\{(a_n, b_n)\}$ are countable, (2.2.1) holds $\forall n$ we get,

$$\mathbb{E}((c(X)|\mathcal{G}) \geq \sup_n (a_n \mathbb{E}(X|\mathcal{G}) + b_n) \implies \mathbb{E}((c(X)|\mathcal{G}) \geq c(\mathbb{E}(X|\mathcal{G}))$$

Letting $c(X) = |X|^p$, we are getting $\mathbb{E}(|X|^p|\mathcal{G}) \geq |\mathbb{E}(X|\mathcal{G})|^p$

Thus by (a),

$$\mathbb{E}(\mathbb{E}(|X|^p|\mathcal{G})) \geq \mathbb{E}(|\mathbb{E}(X|\mathcal{G})|^p) \implies \|\mathbb{E}(X|\mathcal{G})\|_p \leq \|X\|_p$$

- i) Given $\mathcal{H} \subseteq \mathcal{G}$, if $C \in \mathcal{H}$, then

$$\begin{aligned} \int_C \mathbb{E}(X|\mathcal{H}) d\mathbb{P} &= \int_C X d\mathbb{P} = \int_C \mathbb{E}(X|\mathcal{G}) d\mathbb{P} \quad [\text{as } C \in \mathcal{G}] \\ &= \int_C \mathbb{E}(\mathbb{E}(X|\mathcal{G})|\mathcal{H}) d\mathbb{P} \end{aligned}$$

Thus, $\mathbb{E}(\mathbb{E}(X|\mathcal{G})|\mathcal{H}) = \mathbb{E}(X|\mathcal{H})$ a.e.

j) Let $Z = I_B$ where $B \in \mathcal{G}$. Now $C \in \mathcal{G}$,

$$\int_C ZX d\mathbb{P} = \int_{B \cap C} X d\mathbb{P} = \int_{B \cap C} \mathbb{E}(X|\mathcal{G}) d\mathbb{P} = \int_C Z \mathbb{E}(X|\mathcal{G}) d\mathbb{P}$$

Let Z be simple function, say $Z = \sum_{j=1}^n Z_j I_{B_j}$ where $B_j \in \mathcal{G}$.

By linearity of expectation,

$$\mathbb{E}(ZX|\mathcal{G}) = \sum_{j=1}^n Z_j \mathbb{E}(X I_{B_j}|\mathcal{G}) = \sum_{j=1}^n Z_j I_{B_j} \mathbb{E}(X|\mathcal{G}) = Z \mathbb{E}(X|\mathcal{G})$$

Now Z be any $\mathcal{G}$ -measurable function.

Let $Z_1, Z_2, \dots$ be simple and $\mathcal{G}$ -measurable with $|Z_n| \leq |Z|$ and $Z_n \rightarrow Z$.

Now $\mathbb{E}(X Z_n|\mathcal{G}) \rightarrow \mathbb{E}(X Z|\mathcal{G})$ as $X Z$ is integrable. Similarly, $Z_n \mathbb{E}(X|\mathcal{G}) \rightarrow Z \mathbb{E}(X|\mathcal{G})$ as X is integrable. Now,

$$\mathbb{E}(X Z_n|\mathcal{G}) = Z_n \mathbb{E}(X|\mathcal{G}) \quad \text{so, } \mathbb{E}(X Z|\mathcal{G}) = Z \mathbb{E}(X|\mathcal{G}) \quad [\text{as both are finite}]$$

k) We can assume that $X \geq 0$ and $\mathbb{E}(X) < \infty$

For $G \in \mathcal{G}$ and $H \in \mathcal{H}$, $X I_G$ and H are independent.

$$\mathbb{E}(X; G \cap H) = \mathbb{E}((X I_G) I_H) = \mathbb{E}(X I_G) \mathbb{P}(H)$$

If Y is a version $\mathbb{E}(X|\mathcal{G})$. Since Y is $\mathcal{G}$ -measurable, $Y I_G$ is independent of H .

$\mathbb{E}((Y I_G) I_H) = \mathbb{E}(Y I_G) \mathbb{P}(H)$ and we have, $\mathbb{E}(X; G \cap H) = \mathbb{E}(Y; G \cap H)$

□

Chapter 3

Martingales

3.1 Introduction

Here we will focus on the properties as well as the construction of new martingales from pre-existing ones.

The main reference for this chapter is [2, Chapter 10].

3.2 Basic Definitions

3.2.1 Filtered spaces

Filtered spaces is denoted by $(\Omega, \mathcal{F}, \{\mathcal{F}_n\}, \mathbb{P})$. $\{\mathcal{F}_n : n \geq 0\}$ is a filtration i.e. increasing family of sub σ -algebras of $\mathcal{F}$. Thus $\mathcal{F}_n \subseteq \mathcal{F}_{n+1} \quad \forall \quad n$ and $\mathcal{F}_\infty := \sigma(\cup_n \mathcal{F}_n) \subseteq \mathcal{F}$

Idea: The information about ω in Ω available at time n consists precisely of the values of $Z(\omega) \forall \mathcal{F}_n$ -measurable functions Z . Usually $\{\mathcal{F}_n\}$ is the natural filtration of some (stochastic) process $W = \{W_n : n \in \mathbb{Z}^+\}$ and the information about ω at time n consists of $W_0(\omega), W_1(\omega), \dots, W_n(\omega)$. Thus $\mathcal{F}_n = \sigma(W_0, \dots, W_n)$ is the σ -field generated by $W_0, \dots, W_n$

3.2.2 Adapted Process

A process $X = \{X_n : n \geq 0\}$ is called adapted (to the filtration $\{\mathcal{F}_n\}$) if $\forall n, X_n$ is $\mathcal{F}_n$ -measurable.

Idea: If X is adapted, the value $X_n(\omega)$ is known to us at time n . Usually, $\mathcal{F}_n = \sigma(W_0, \dots, W_n)$ and $X_n = f_n(W_0, \dots, W_n)$ for some f_n which is $\mathcal{B}^{n+1}$ measurable function on $\mathbb{R}^{n+1}$

3.3 Martingale , Supermartingale and Submartingale

Definition 3.3.1. A process X is called a martingale (wrt $\mathcal{F}_n, \mathbb{P}$) if

- i) X is adapted
- ii) $\mathbb{E}(|X_n|) < \infty \quad \forall n$
- iii) $\mathbb{E}(X_n | \mathcal{F}_{n-1}) = X_{n-1}$ a.s. $\forall n \geq 1$ (i.e. $\mathbb{E}((\mathbb{E}(X_n | \mathcal{F}_{n-1}))) = \mathbb{E}(X_{n-1}) \implies \mathbb{E}(X_n) = \mathbb{E}(X_{n-1})$ a.s)

Supermartingale is defined similarly except that (iii) is replaced by

$$\mathbb{E}(X_n | \mathcal{F}_{n-1}) \leq X_{n-1} \quad \text{a.s.} \quad \forall \quad n \geq 1$$

Submartingale is defined similarly except that (iii) is replaced by

$$\mathbb{E}(X_n | \mathcal{F}_{n-1}) \geq X_{n-1} \quad \text{a.s.} \quad \forall \quad n \geq 1$$

Remark 3.3.2. X is a supermartingale iff $-X$ is an submartingale . X is a martingale iff X is both supermartingale and submartingale.

Example 3.3.3. Suppose $\{X_n\}_{n \geq 1}$ be an i.i.d sequence where $X_n = \begin{cases} +1 & \text{with } p \\ -1 & \text{with } 1-p \end{cases}$ Let $Y_0 := 0$, $Y_n := X_1 + \dots + X_n$, $\mathcal{F}_n := \sigma(X_1, \dots, X_n)$

Claim: $\{Y_n\}_n$ is a martingale

Clearly Y is adapted and $\mathbb{E}(|Y_n|) < \infty \quad \forall n$ as $\mathbb{E}(|X_n|) < \infty \quad \forall n$.

Need to check $\mathbb{E}(Y_n | \mathcal{F}_{n-1}) = Y_{n-1}$ a.s.

Now R.H.S. $= Y_{n-1} = \mathbb{E}(Y_{n-1} | \mathcal{F}_{n-1})$ a.s.

Thus it is sufficient to check $\mathbb{E}(Y_n - Y_{n-1} | \mathcal{F}_{n-1}) = 0$ a.s.

Thus now, L.H.S. $= \mathbb{E}(X_n | \mathcal{F}_{n-1}) = \mathbb{E}(X_n) [\text{as } \mathcal{F}_{n-1} := \sigma(X_1, \dots, X_{n-1})]$ and thus is equal to $2p - 1$

Thus $\{Y_n\}$ as defined above is a martingale iff $2p - 1 = 0 \Leftrightarrow p = \frac{1}{2}$

Similarly, if $p > \frac{1}{2}$,

$$\mathbb{E}(Y_n - Y_{n-1} | \mathcal{F}_{n-1}) = 2p - 1 > 0 \quad \text{a.s.} \quad \mathbb{E}(Y_n | \mathcal{F}_{n-1}) > Y_{n-1} \quad \text{a.s.} \quad (\text{Submartingale})$$

Again, if $p < \frac{1}{2}$,

$$\mathbb{E}(Y_n - Y_{n-1} | \mathcal{F}_{n-1}) = 2p - 1 < 0 \quad \text{a.s.} \quad \mathbb{E}(Y_n | \mathcal{F}_{n-1}) < Y_{n-1} \quad \text{a.s.} \quad (\text{Submartingale})$$

Remark 3.3.4. If X is a supermartingale, for $m < n$, $\mathcal{F}_m \subseteq \mathcal{F}_{n-1}$, thus

$$\mathbb{E}(X_n | \mathcal{F}_m) = \mathbb{E}(\mathbb{E}(X_n | \mathcal{F}_{n-1}) | \mathcal{F}_m) \leq \mathbb{E}(X_{n-1} | \mathcal{F}_m) \dots \leq X_m \quad \text{a.s.}$$

Example 3.3.5. a) **Sums of independent zero-mean r.v.s**

$X_1, X_2, \dots$ sequence of independent r.v.s with $\mathbb{E}(|X_k|) < \infty \quad \forall k$ and $\mathbb{E}(X_k) = 0 \quad \forall k$.

Define

$$S_0 := 0, \quad S_n := X_1 + \dots + X_n, \quad \mathcal{F}_0 := (\phi, \Omega), \quad \mathcal{F}_n := \sigma(X_1, \dots, X_n)$$

So, $\forall n$ we have,

$$\begin{aligned} \mathbb{E}(S_n | \mathcal{F}_{n-1}) &= \mathbb{E}(S_{n-1} | \mathcal{F}_{n-1}) + \mathbb{E}(X_n | \mathcal{F}_{n-1}) \quad (\text{By linearity}) \\ &= S_{n-1} + \mathbb{E}(X_n) \quad (\because S_{n-1} \text{ is } \mathcal{F}_{n-1} \text{ measurable and } X_n \text{ is independent of } \mathcal{F}_{n-1}) \\ &= S_{n-1} \quad (\text{a.s.}) [\mathbb{E}(X_n) = 0] \end{aligned}$$

Thus S is a martingale.

b) **Product of non-negative independent r.v.s with mean 1**

$X_1, X_2, \dots$ sequence of non- negative independent r.v.s with $\mathbb{E}(X_k) = 1 \quad \forall k$ and $\mathbb{E}(X_k) = 0 \quad \forall k$.

Define

$$M_0 := 1, \quad M_n := X_1 X_2 \dots X_n, \quad \mathcal{F}_0 := (\phi, \Omega), \quad \mathcal{F}_n := \sigma(X_1, \dots, X_n)$$

So, $\forall n$ we have,

$$\begin{aligned} \mathbb{E}(M_n | \mathcal{F}_{n-1}) &= M_{n-1} \mathbb{E}(X_n) \quad (\text{as } M_{n-1} \text{ is } \mathcal{F}_{n-1} \text{ measurable and } X_n \text{ is independent of } \mathcal{F}_{n-1}) \\ &= M_{n-1} \end{aligned}$$

Thus M is a martingale.

c) **Accumulating data about a r.v**

Let $\{\mathcal{F}\}$ be our filtration and $\xi \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$

Define $M_n := \mathbb{E}(\mathbb{E}(\xi | \mathcal{F}_n) | \mathcal{F}_{n-1}) = M_{n-1} \quad [\because \mathcal{F}_{n-1} \subseteq \mathcal{F}_n]$

Hence M is a martingale.

3.3.1 Previsible process

We call a process $C = \{C_n : n \in \mathbb{N}\}$ previsible if C_n is $\mathcal{F}_{n-1}$ measurable. ($n \geq 1$).

Note: C has parameter set $\mathbb{N}$ rather than $\mathbb{Z}^+$. C_0 doesn't exist.

Proposition 3.3.6 (Fundamental Principle[2, 10.7]). (i) Let C be a bounded, non-negative, previsible process so that for some $K \in [0, \infty)$, $|C_n(\omega)| \leq K \forall n, \omega$. Let X be a martingale (supermartingale) implying $(C \circ X)$ is a martingale (supermartingale) null at 0.

(ii) If C is a bounded previsible process and X is a martingale, then $C \circ X$ is a martingale null at 0.

(iii) In (i) and (ii), the bounded condition of C may be replaced by the condition $C_n \in \mathcal{L}^2 \forall n$, provided $X_n \in \mathcal{L}^2 \forall n$.

Proof. i) Write Y for $C \circ X$.

Now C_n is bounded non-negative and $\mathcal{F}_{n-1}$ measurable. Thus,

$$\mathbb{E}(Y_n - Y_{n-1} | \mathcal{F}_{n-1}) = C_n \mathbb{E}(X_n - X_{n-1} | \mathcal{F}_{n-1}) = 0 (\leq 0)$$

ii) It is evident for the case of martingale from the previous one. [Not applicable for supermartingale].

iii) If $C_n \in \mathcal{L}^2$, then $|C_n(\omega)| \leq M$ where $M \in [0, \infty)$.

If $X_n \in \mathcal{L}^2$, then $|(X_n - X_{n-1})(\omega)|$ is bounded in $\mathcal{L}^1 \forall n, \omega$.

□

3.3.2 Stopping Time

Definition 3.3.7. A map $T : \Omega \rightarrow \{0, 1, 2, \dots, \infty\}$ is called a stopping time if

$$\{T \leq n\} = \{\omega : T(\omega) \leq n\} \in \mathcal{F}_n \forall n \leq \infty \quad (3.3.1)$$

Equivalently,

$$\{T = n\} = \{\omega : T(\omega) = n\} \in \mathcal{F}_n \forall n \leq \infty \quad (3.3.2)$$

Note that T can be ∞ .

Proof of the equivalence. Suppose T has the property (3.3.1), $\{T = 0\}$ and $\{T \leq 0\}$ present the same set. Thus $\{T = 0\} \in \mathcal{F}_0$.

Now,

$$\{T \leq n-1\} \in \mathcal{F}_{n-1} \implies \{T \leq n-1\}^c \in \mathcal{F}_{n-1} \implies \{T \leq n-1\}^c \in \mathcal{F}_n [\text{as } \mathcal{F}_{n-1} \subseteq \mathcal{F}_n].$$

Thus, $\{T = n\} = \{T \leq n\} \cap \{T \leq n-1\}^c \in \mathcal{F}_n \forall n < \infty$.

Now $\{T = n\} \in \mathcal{F}_\infty$ [as $\mathcal{F}_n \subseteq \mathcal{F}_\infty \forall n$].

Thus $\cup_n \{T = n\} \in \mathcal{F}_\infty \implies \{T = \infty\} = (\cup_n \{T = n\})^c \in \mathcal{F}_\infty$

Suppose T has the property (3.3.2),

$$\{T \leq n\} = \cup_{k=0}^n \{T = k\} \in \mathcal{F}_n [\text{as each set } \in \mathcal{F}_n]$$

□

Idea : T is a time when you can decide to stop playing our game. Whether you stop or not totally depends on the information upto the n -th game (included): $\{T = n\} \in \mathcal{F}_n$

Example 3.3.8. Suppose $\{A_n\}$ is an adapted process and that $B \in \mathcal{B}$. Let $T = \inf\{n \geq 0 : A_n \in B\}$ = time of first entry of A_n into set B . $\inf(\emptyset) = \infty$, so that $T = \infty$ if A never enters set B . $\{T \leq n\} = \cup_{k \leq n} \{A_k \in B\} \in \mathcal{F}_n$ so that T is a stopping time.

3.3.3 Stopped martingales are supermartingales

Let X be a martingale and T be stopping time. Let us assume that we always bet 1 unit and quit playing at time T . Then the ‘stake process’ is $C^{(T)}$ where $\forall n \in \mathcal{N}$.

$C_n^{(T)} = I_{\{n \leq T\}}$ so $C_n^{(T)}(\omega) = \begin{cases} 1 & n \leq T(\omega) \\ 0 & \text{otherwise} \end{cases}$ The ‘winnings process’ is the process with value at time n equal to

$$(C^{(T)} \circ X)_n = X_{T \wedge n} - X_0$$

If X_T denotes the process X stopped at T . Then, $X_n^T(\omega) = X_{T(\omega) \wedge n}(\omega)$ which means that $C^{(T)} \circ X = X^T - X_0$.

$C^{(T)}$ is clearly bounded by 1 and non-negative.

Moreover, $C^{(T)}$ can only be 0 and 1. Now, $\{C^{(T)=0}\} = \{T \leq n-1\} \in \mathcal{F}_{n-1}$ and $\{C^{(T)=1}\} = \{T \geq n\} = \{T \leq n-1\}^c \in \mathcal{F}_{n-1}$ implying that $C^{(T)}$ is $\mathcal{F}_{n-1}$ measurable.

Thus using (i) from the fundamental principle we get the result.

Theorem 3.3.9 ([2, 10.9]). (i) If X is a supermartingale and T is a stopping time, then the stopped process $X^T = \{X_{T \wedge n} : n \in \mathbb{Z}^+\}$ is a supermartingale. In particular, $\mathbb{E}(X_{T \wedge n}) \leq \mathbb{E}(X_0) \forall n$.

(ii) If X is a martingale and T is a stopped time, then X^T is a martingale. In particular, $\mathbb{E}(X_{T \wedge n}) = \mathbb{E}(X_0) \forall n$

Theorem 3.3.10 (Doob’s Optional Stopping Theorem [2, 10.10]). a) Let T be a stopping time and X be a supermartingale. Then X_T is integrable and $\mathbb{E}(X_T) \leq \mathbb{E}(X_0)$ in each of the following situations

- i) T is bounded.
- ii) X is bounded and T is finite a.e.
- iii) $\mathbb{E}(T) < \infty$ and for some K in $\mathbb{R}^+$, $|X_n(\omega) - X_{n-1}(\omega)| \leq K \forall \omega, n$.

b) If any of the conditions (i)-(iii) holds and X is a martingale, then $\mathbb{E}(X_T) = \mathbb{E}(X_0)$

Proof. a) From theorem 3.3.9, we know that $X_{T \wedge n}$ is integrable and $\mathbb{E}(X_{T \wedge n} - X_0) \leq 0$.

- i) T is bounded. Thus $T(\omega) \leq N \forall \omega$ and then, $X_{T \wedge n} = X_T \forall n > N$. Now $\mathbb{E}(X_{T \wedge n} - X_0) \leq 0 \forall n$ leads to $\mathbb{E}(X_T) \leq \mathbb{E}(X_0) \forall n > N$
- ii) Here $|X_n(\omega)| \leq K \forall (n, \omega)$. Now $\mathbb{E}(X_{T \wedge n} - X_0) \leq 0 \forall n$ and $X_{T \wedge n} - X_0 \rightarrow X_T - X_0$ as T is finite a.s. Thus $\mathbb{E}(X_T - X_0) \leq 0$
- iii) $|X_{T \wedge n} - X_0| = |\sum_{k=1}^{T \wedge n} (X_k - X_{k-1})| \leq K.T$ for some $K \in \mathbb{R}^+$ and $\mathbb{E}(KT) < \infty$. Thus using D.C.T, we get $\lim_{n \rightarrow \infty} \mathbb{E}(X_{T \wedge n} - X_0) = \mathbb{E}(X_T - X_0) \leq 0$

b) Applying (a) to X and $-X$ we can get the results. □

Corollary 3.3.11. c) Suppose that M is a martingale with $|M_n - M_{n-1}| \leq K_1$ and C is a pre-visible process bounded by K_2 and that T is a stopping time such that $\mathbb{E}(T) < \infty$. Then $\mathbb{E}(C \circ M)_T = 0$

d) If X is a non-negative supermartingale and T is a stopping time which is a.s. finite, then $\mathbb{E}(X_T) \leq \mathbb{E}(X_0)$.

Proof. c) Following are the conditions that we are provided with

- M is a martingale which implies $\mathbb{E}(M_n | \mathcal{F}_{n-1}) = M_{n-1}$ a.s. ($n \geq 1$).

- $|M_n - M_{n-1}| \leq K_1 \forall (n, \omega)$.
- C_n is $\mathcal{F}_{n-1}$ measurable and $|C_n(\omega)| \leq K_2 \forall n$.
- T is a stopping time with $\mathbb{E}(T) < \infty$

We write $C \circ M$ as Y . Now, $\mathbb{E}(Y_n - Y_{n-1} | \mathcal{F}_{n-1}) = c_n \mathbb{E}(M_n - M_{n-1} | \mathcal{F}_{n-1}) = 0$

Thus, $|Y_{T \wedge n} - Y_0| = |\sum_{k=1}^{T \wedge n} (Y_k - Y_{k-1})| = |\sum_{k=1}^{T \wedge n} c_k (M_k - M_{k-1})| \leq K_2 K_1 T$.

This uniform bound required for applying DCT.

Now,

$$\mathbb{E}(Y_{T \wedge n} - Y_0) = c_1 \mathbb{E}(M_1 - M_0 | \mathcal{F}_0) + c_2 \mathbb{E}(M_2 - M_1 | \mathcal{F}_1) + \dots + c_{T \wedge n} \mathbb{E}(M_{T \wedge n} - M_{T \wedge n-1} | \mathcal{F}_{T \wedge n-1}) = 0$$

Again, $Y_{T \wedge n} - Y_0 \rightarrow Y_T - Y_0$ and hence, $\mathbb{E}(Y_{T \wedge n} - Y_0) \rightarrow \mathbb{E}(Y_T - Y_0)$. But $\mathbb{E}(Y_{T \wedge n} - Y_0) = 0 \forall n$ which means that $\mathbb{E}(Y_T - Y_0) = 0$ and finally $\mathbb{E}(Y_T) = \mathbb{E}(Y_0)$

d) $X_{T \wedge n} \xrightarrow{pt.wise} X_T$ as T is finite a.s.

Using Fatou's lemma ,

$$\mathbb{E}(X_T | \mathcal{F}_0) = \mathbb{E}(\liminf X_{T \wedge n} | \mathcal{F}_0) \leq \liminf \mathbb{E}(X_{T \wedge n} | \mathcal{F}_0) \leq X_0 \quad \text{a.s.}$$

So $\mathbb{E}(X_T | \mathcal{F}_0) \leq X_0$ a.s., implying $\mathbb{E}(X_T) \leq \mathbb{E}(X_0)$ a.s. □

Lemma 3.3.12 ([2, 10.11]). *Suppose that T is stopping time such that for some N in $\mathbb{N}$ and for some $\epsilon > 0$, we have , for every n in $\mathbb{N}$: $\mathbb{P}(T \leq n + N | \mathcal{F}_n) > \epsilon$ a.s. Then $\mathbb{E}(T) < \infty$*

Proof. Since, $\mathbb{P}(T \leq n + N | \mathcal{F}_n) > \epsilon$ we have $\mathbb{E}(1_{T \leq n+N} | \mathcal{F}_n) > \epsilon$ a.s. or $\mathbb{E}(1_{T > n+N} | \mathcal{F}_n) < 1 - \epsilon$ a.s.. By definition of conditional expectation, we have $\mathbb{P}((T > n + N) \cap A) \leq (1 - \epsilon) \mathbb{P}(A) \forall A \in \mathcal{F}_n$.

Now take $n = (k - 1)N$; $A = (T > (k - 1)N)$, thus this implies that

$$\mathbb{P}(T > kN) \leq (1 - \epsilon) \mathbb{P}(T > (k - 1)N) \dots \leq (1 - \epsilon)^k$$

Thus $\mathbb{E}(T) = \sum_{k=1}^{\infty} kN \mathbb{P}(T > kN) \leq N \sum_{k=1}^{\infty} k(1 - \epsilon)^k < \infty$. This completes the proof. □

3.3.4 Hitting times for simple random walk

$\{X_n : n \in \mathbb{N}\}$ is a sequence of i.i.d. r.v.s and $X_n \sim X$ where $\mathbb{P}(X = 1) = \frac{1}{2} = \mathbb{P}(X = -1)$

$S_0 := 0$; $S_n := X_1 + X_2 + \dots + X_n$ and set $T := \inf\{n : S_n = 1\}$.

We look at the set $\{T = k\} = \{S_k = 1\} \in \mathcal{F}_k = \sigma(X_0, \dots, X_n) = \sigma(S_0, \dots, S_n)$.

So S_n is adapted and T is a stopping time .

For $\theta \in \mathbb{R}$,

$$\mathbb{E}(e^{(\theta X)}) = \frac{1}{2}(e^\theta + e^{-\theta}) = \cosh \theta \implies \mathbb{E}[(\text{sech } \theta) e^{(\theta X_n)}] = 1 \forall n$$

Let $M_n^\theta = (\text{sech } \theta)^n e^{(\theta S_n)}$. Clearly , M_n is $\mathcal{F}_n$ -measurable.

$$\begin{aligned} \mathbb{E}(M_n^\theta | \mathcal{F}_{n-1}) &= \mathbb{E}((\text{sech } \theta) e^{\theta X_n} (\text{sech } \theta)^{n-1} e^{\theta S_{n-1}} | \mathcal{F}_{n-1}) = M_{n-1}^\theta \mathbb{E}[(\text{sech } \theta) e^{(\theta X_n)} | \mathcal{F}_{n-1}] \\ &= M_{n-1}^\theta \mathbb{E}[(\text{sech } \theta) e^{(\theta X_n)}] = M_{n-1}^\theta \end{aligned}$$

Therefore, M^θ is a martingale.

Now , $\mathbb{E}(M_{T \wedge n}^\theta) = \mathbb{E}[(\text{sech } \theta)^{T \wedge n} e^{\theta S_{T \wedge n}}] = 1 \quad \forall n$

We insist that $\theta > 0$, then $\exp(\theta S_{T \wedge n}) \leq e^\theta \implies M_{T \wedge n}^\theta \leq e^\theta$

As $n \rightarrow \infty$, $M_{T \wedge n}^\theta \rightarrow M_T^\theta (= 0 \text{ if } T = \infty)$

By Bounded Convergence theorem ,

$$\mathbb{E}(M_T^\theta) = 1 = \mathbb{E}((\text{sech } \theta)^T e^\theta)$$

$$\implies \mathbb{E}((\operatorname{sech} \theta))^T = e^{-\theta} \quad \forall \quad \theta > 0$$

Let $\theta \downarrow 0$, $(\operatorname{sech} \theta)^T \uparrow 1$ if $T < \infty$ and $(\operatorname{sech} \theta)^T \uparrow 0$ if $T = \infty$.

$$\begin{aligned} \mathbb{E}((\operatorname{sech} \theta))^T &= \mathbb{E}((\operatorname{sech} \theta))^T I_{(T < \infty)} + \mathbb{E}((\operatorname{sech} \theta))^T I_{(T = \infty)} = e^{-\theta} \\ \mathbb{E}(I_{(T < \infty)}) &= 1 \implies \mathbb{P}(T < \infty) = 1 \end{aligned}$$

Putting $\alpha = \operatorname{sech} \theta$ we get, $\mathbb{E}(\alpha^T) = \sum \alpha^n \mathbb{P}(T = n) = e^{-\theta} = \frac{1 - \sqrt{1 - \alpha^2}}{\alpha}$

Chapter 4

Martingale Convergence Theorem

4.1 Introduction

In this chapter, we discuss the conditions for the existence of X_∞ in the martingale setup that has been built up in the previous chapters.

The main reference for this chapter is [2, Chapter 11].

4.2 Upcrossings

Definition 4.2.1. The number $U_N[a, b](\omega)$ of upcrossings of $[a, b]$ made by $n \mapsto X_n(\omega)$ by the time N is defined to be the largest k in $\mathbb{Z}^+$ such that we can find

$$0 \leq s_1 < t_1 < s_2 < t_2 \dots < s_k < t_k \leq N$$

with $X_{s_i}(\omega) < a; X_{t_i}(\omega) > b$ ($1 \leq i \leq k$)

Now $Y_0(\omega) := 0$

$$Y_N(\omega) \geq (b - a)U_N[a, b](\omega) - [X_N(\omega) - a]^-$$

Every upcrossing of $[a, b]$ increases the Y -value by at least $(b - a)$ while the $[X_N(\omega) - a]^-$ overemphasizes the loss during the last 'interval of play'.

Lemma 4.2.2 (Doob's Upcrossing Lemma [2, 11.3]). *Let X be a supermartingale and $U_N[a, b]$ be the number of upcrossings of $[a, b]$ by the time N . Then*

$$(b - a)\mathbb{E}(U_N[a, b]) \leq \mathbb{E}[(X_N - a)^-]$$

Proof. Define

$$C_1 := I_{\{X_0 < a\}} \\ \text{and for } n \geq 2, \quad C_n := I_{\{C_{n-1}=1\}}I_{\{X_{n-1} \leq b\}} + I_{\{C_{n-1}=0\}}I_{\{X_{n-1} < a\}}.$$

Clearly C is a bounded, previsible and non-negative.

Thus X being a supermartingale, implies Y to be a supermartingale too.

Now,

$$\mathbb{E}(Y_n) = \mathbb{E}\left(\sum_{k=1}^N C_k(X_k - X_{k-1})\right) = \sum_{k=1}^N C_k \mathbb{E}(X_k - X_{k-1}) \leq 0$$

Using the previous inequality we get,

$$(b - a)\mathbb{E}(U_N[a, b]) \leq \mathbb{E}[(X_N - a)^-]$$

□

Corollary 4.2.3. *[[2, 11.4]] X is a supermartingale bounded in $\mathcal{L}^1$ i.e. $\sup_n \mathbb{E}|X_n| < \infty$. Let $a, b \in \mathbb{R}$ with $a < b$.*

Then with $U_\infty[a, b] := \uparrow \lim_N U_N[a, b]$,

$$(b - a)\mathbb{E}(U_\infty[a, b]) \leq |a| + \sup_N \mathbb{E}(|X_n|) < \infty$$

so that $\mathbb{P}[U_\infty[a, b] = \infty] = 0$ i.e. there can be only finitely many upcrossing.

Proof. Using the above lemma 4.2.2,

$$(b - a)\mathbb{E}(U_N[a, b]) \leq \mathbb{E}[(X_n - a)^-] \leq |a| + \mathbb{E}|X_n| \leq |a| + \sup_N \mathbb{E}(|X_n|)$$

As $U_N[a, b]$ is non-negative by definition, by MCT we get $(b - a)\mathbb{E}(U_\infty[a, b]) < \infty$.

$$\mathbb{E}(U_\infty[a, b]) \text{ is finite implying } \mathbb{P}(U_\infty[a, b] = \infty) = 0$$

□

Theorem 4.2.4 (Doob's 'Forward' Convergence Theorem[2, 11.5]). *Let X be supermartingale bounded in $\mathcal{L}^1$ i.e. $\sup_n \mathbb{E}|X_n| < \infty$. Then a.s. $X_\infty := \lim X_n$ exists and is finite. For definiteness, we define $X_\infty := \limsup X_n(\omega) \quad \forall \omega$ so that X_∞ is $\mathcal{F}_\infty$ measurable and $X_\infty := \lim X_n$ a.s.*

Proof.

$$\begin{aligned} \Lambda &:= \{\omega : X_n(\omega) \text{ does not converge to a limit in } [-\infty, \infty]\} \\ &= \{\omega : \liminf X_n(\omega) < \limsup X_n(\omega)\} \\ &= \cup_{\{a, b \in \mathbb{Q} : a < b\}} \{\omega : \liminf X_n(\omega) < a < b < \limsup X_n(\omega)\} \\ &=: \cup \Lambda_{a, b} (\text{say}) \end{aligned}$$

But $\Lambda_{a, b} \subseteq \{\omega : (U_\infty[a, b] = \infty)\}$.

Now from the previous corollary 4.2.3, $\mathbb{P}(U_\infty[a, b] = \infty) = 0$ which implies $\mathbb{P}(\Lambda_{a, b}) = 0$.

As Λ is a countable union of sets $\Lambda_{a, b}$, $\mathbb{P}(\Lambda) = 0$.

Thus $X_\infty := \lim X_n$ exists a.s. in $[-\infty, \infty]$.

By Fatou's Lemma,

$$\mathbb{E}|X_\infty| = \mathbb{E}(\liminf |X_n|) \leq \liminf \mathbb{E}|X_n| \leq \sup \mathbb{E}|X_n| < \infty$$

so that X_∞ is finite a.s.

□

Note 4.2.5. It need not be true that $X_n \rightarrow X_\infty$ in $\mathcal{L}^1$

Corollary 4.2.6. *If X is non-negative super martingale, then $X_\infty := \lim X_n$ exists a.s.*

Proof. Clearly $\mathbb{E}(X_n) \leq \mathbb{E}(X_0) \quad \forall n$. Now the same argument as before is to be used.

□

Chapter 5

Martingales bounded in $\mathcal{L}^2$

5.1 Introduction

We discuss martingales that are bounded in $\mathcal{L}^2$ space and look into their properties and their connection with the previous chapters.

The main reference for this chapter is [2, Chapter 12].

5.2 Martingales in $\mathcal{L}^2$; orthogonality of increments

Let $M = (M_n : n \geq 0)$ be a martingale in $\mathcal{L}^2$ i.e each M_n is in $\mathcal{L}^2$ so that $\mathbb{E}(M_n^2) < \infty \quad \forall n$. Then for $s, t, u, v \in \mathbb{Z}^+$ with $s \leq t \leq u \leq v$,

$$\mathbb{E}(M_v | \mathcal{F}_u) = M_u \quad (\text{a.s.})$$

so that $M_v - M_u$ is orthogonal to $\mathcal{L}^2(\mathcal{F}_u)$ and in particular, $\langle M_t - M_s, M_v - M_u \rangle = 0$.

$$\mathbb{E}(M_v - M_u | \mathcal{F}_u) = 0 \implies \mathbb{E}((M_v - M_u)I_B) = 0 \quad \forall B \in \mathcal{F}_u$$

Now there exists simple $X_n \rightarrow X$ s.t. $|X_n| < |X|$

$$(\mathbb{E}(M_v - M_u)X_n)^2 \leq \mathbb{E}(M_v - M_u)^2 \mathbb{E}(X_n^2) \leq \mathbb{E}(M_v - M_u)^2 \mathbb{E}(X^2) < \infty$$

Using DCT, $\mathbb{E}(M_v - M_u)X = 0 \quad \forall X$ that are $\mathcal{F}_u$ -measurable

Now, clearly $(M_t - M_s)$ is $\mathcal{F}_u$ -measurable

Hence the formula $M_n = M_0 + \sum_{k=1}^n (M_k - M_{k-1})$ expresses M_n as the sum of orthogonal terms and Pythagoras theorem yields

$$\mathbb{E}(M_n^2) = \mathbb{E}(M_0^2) + \sum_{k=1}^n \mathbb{E}(M_k - M_{k-1})^2 \quad (5.2.1)$$

Theorem 5.2.1 ([2, 12.1]). *M be a martingale for which $M_n \in \mathcal{L}^2 \quad \forall n$. Then M is bounded in $\mathcal{L}^2$ iff*

$$\sum \mathbb{E}(M_k - M_{k-1})^2 < \infty \quad (5.2.2)$$

and this obtains $M_n \rightarrow M_\infty$ almost surely and in $\mathcal{L}^2$.

Proof. From (5.2.1) it is obvious that $\sum \mathbb{E}(M_k - M_{k-1})^2 < \infty \Leftrightarrow M$ is bounded in $\mathcal{L}^2$

Now that (5.2.2) holds, M is bounded in $\mathcal{L}^2 \implies M$ is bounded in $\mathcal{L}^1$.

Thus, by Doob's 'Forward' Convergence Theorem, $M_\infty := \lim M_n$ exists a.s.

$$\therefore \mathbb{E}(M_{n+r} - M_n)^2 = \sum_{k=n+1}^{n+r} \mathbb{E}(M_k - M_{k-1})^2 \quad (5.2.3)$$

Letting $r \rightarrow \infty$ and applying Fataou's lemma ,

$$\begin{aligned} \mathbb{E}((\lim_{r \rightarrow \infty} (M_{n+r} - M_n))^2) &= \mathbb{E}(\liminf (M_{n+r} - M_n))^2 \leq \liminf \sum_{k=n+1}^{n+r} \mathbb{E}(M_k - M_{k-1})^2 \\ \implies \mathbb{E}(M_\infty - M_n)^2 &\leq \sum_{k \geq n+1} \mathbb{E}(M_k - M_{k-1})^2 \end{aligned} \quad (5.2.4)$$

Hence ,

$$\lim_n \mathbb{E}(M_\infty - M_n)^2 = 0 \quad (5.2.5)$$

so that $M_n \rightarrow M_\infty$ in $\mathcal{L}^2$.

Also (5.3.1) allows us to deduce from (5.2.3) that (5.2.4) holds with equality. $\square$

5.3 Sums of zero- mean independent variables in $\mathcal{L}^2$

Theorem 5.3.1 ([2, 12.2]). *Suppose that $\{X_k; k \in \mathbb{N}\}$ is a sequence of independent r.v.s such that $\forall k \ \mathbb{E}(X_k) = 0, \sigma_k^2 := \text{Var}(X_k) < \infty$*

a) *Then , $\sum \sigma_k^2 < \infty \implies \sum X_k$ converges a.s.*

b) *If $|X_k(\omega)| \leq K, K \in [0, \infty) \forall \ k, \omega$, then $\sum X_k$ converges a.s. $\implies \sum \sigma_k^2 < \infty$*

Proof. Define

$$\begin{aligned} \mathcal{F}_n &:= \sigma(X_1, \dots, X_n), \quad M_n := X_1 + X_2 + \dots + X_n, \\ A_n &:= \sum_{k=1}^n \sigma_k^2, \quad N_n := M_n^2 - A_n, \\ \mathcal{F}_0 &:= \{\phi, \Omega\}, \quad M_0 := 0, \quad A_0 := 0, \quad N_0 := 0 \end{aligned}$$

[M^2 is a submartingale applying Jensen Inequality]

a) Now M is a martingale.

$$\mathbb{E}(M_k - M_{k-1})^2 = \mathbb{E}(X_k^2) = \sigma_k^2 \quad (5.3.1)$$

Now , $\mathbb{E}(M_n^2) = \sum_{k=1}^n \sigma_k^2 = A_n$

Now , if $\sum \sigma_k^2 < \infty$, then M is bounded in $\mathcal{L}^2$ so that $\lim_n M_n$ exists (Using Doob's CT)

b) We can strengthen (5.3.1) as follows .

$$\mathbb{E}[(M_k - M_{k-1})^2 | \mathcal{F}_{k-1}] = \mathbb{E}(X_k^2 | \mathcal{F}_{k-1}) = \mathbb{E}(X_k^2) = \sigma_k^2$$

$$\begin{aligned} \therefore \sigma_k^2 &= \mathbb{E}(M_k^2 | \mathcal{F}_{k-1}) - 2M_{k-1}\mathbb{E}(M_k | \mathcal{F}_{k-1}) + M_{k-1}^2 \quad \text{a.s.} \\ &= \mathbb{E}(M_k^2 | \mathcal{F}_{k-1}) - M_{k-1}^2 \quad \text{a.s.} \end{aligned}$$

$$\begin{aligned} \implies \sigma_k^2 &= \mathbb{E}(M_k^2 | \mathcal{F}_{k-1}) - \sum_{n=1}^{k-1} \sigma_n^2 - (M_{k-1}^2 - \sum_{n=1}^{k-1} \sigma_n^2) \\ \implies \mathbb{E}(M_k^2 - A_k | \mathcal{F}_{k-1}) &= M_{k-1}^2 - A_{k-1} \\ \implies \mathbb{E}(N_k | \mathcal{F}_{k-1}) &= N_{k-1} \end{aligned}$$

$\therefore N$ is a martingale.

Note that the submartingale M_n^2 is being decomposed into a martingale and something.

Let $c \in (0, \infty)$ and define $T := \inf\{r : |M_r| > c\}$. Now we know , N^T is a martingale.

$$\mathbb{E}(N_n^T) = \mathbb{E}(M_n^T)^2 - \mathbb{E}(A_{T \wedge n}) = 0$$

But since $|M_T - M_{T-1}| = |X_T| \leq K$ if T is finite.

$$|M_n^T| = |M_n^T - M_{n-1}^T + M_{n-1}^T| \leq |M_n^T - M_{n-1}^T| + |M_{n-1}^T| \leq K + c\mathbb{E}(A_{T \wedge n}) \leq (K+c)^2 \quad \forall n \quad (5.3.2)$$

Now, as $\sum X_n$ converges a.s. $\implies$ Partial sums of $\sum X_n$ is bounded a.s.

and it must be the case that for $c, \mathbb{P}(T = \infty) > 0$

From (5.3.2), it is evident,

$$\begin{aligned} \mathbb{E}(A_{T \wedge n} I_{(T < \infty)}) + \mathbb{E}(A_{T \wedge n} I_{(T = \infty)}) &\leq (K+c)^2 \\ \implies \mathbb{E}(A_n I_{(T = \infty)}) &\leq (K+c)^2 \\ \implies \sum_{k=1}^n \sigma_k^2 \mathbb{P}(T = \infty) &\leq (K+c)^2 \quad \forall n \end{aligned}$$

It is now clear from (5.3.2) that $A_\infty := \sum \sigma_k^2 < \infty$

□

Remark 5.3.2. The proof of (b) shows that if $\{X_n\}$ is a sequence of independent zero means r.v.s uniformly bounded by K , then $\mathbb{P}(\text{Partial sums of } \sum X_k \text{ are bounded}) > 0 \implies \sum X_k \text{ converges a.s.}$

Lemma 5.3.3 ([2, 12.4]). Suppose that $\{X_n\}$ is a sequence of independent r.v.s bounded by a constant K in $[0, \infty)$ i.e. $|X_n(\omega)| \leq K \quad \forall n \forall \omega$.

Then if $\sum X_n$ converges a.s. $\implies \sum \mathbb{E}(X_n)$ converges and $\sum \text{Var}(X_n)$ converges.

Theorem 5.3.4 (Kolmogorov's 3 Series theorem [2, 12.5]). Let $\{X_n\}$ be a sequence of independent r.v.s. Then $\sum X_n$ converges a.s. iff for some (then for every) $K > 0$, the following 3 properties hold:

i) $\sum_n \mathbb{P}(|X_n| > K) < \infty$

ii) $\sum_n \mathbb{E}(X_n^K)$ converges

iii) $\sum \text{Var}(X_n^K) < \infty$

where $X_n^K(\omega) := \begin{cases} X_n(\omega) & \text{if } |X_n(\omega)| \leq K \\ 0 & \text{if } |X_n(\omega)| > K \end{cases}$

Lemma 5.3.5 (Cesaro's Lemma [2, 12.6]). Suppose that $\{b_n\}$ is a sequence of strictly positive real numbers with $b_n \uparrow \infty$ and $\{V_n\}$ a convergent sequence $V_n \rightarrow V_\infty \in \mathbb{R}$.

Then $\frac{1}{b_n} \sum_{k=1}^n (b_k - b_{k-1}) V_k \rightarrow V_\infty$ (as $n \rightarrow \infty$). Here, $b_0 := 0$

Lemma 5.3.6 (Kronecker's Lemma [2, 12.7]). Let $\{b_n\}$ be a sequence such that $b_n > 0$ & $b_n \uparrow \infty$. Let $\{X_n\}$ be seq of real numbers and define $S_n := x_1 + \dots + x_n$. Then $\sum \frac{x_n}{b_n}$ converges $\implies \frac{S_n}{b_n} \rightarrow 0$

Lemma 5.3.7 (Strong Law under variance constraints [2, 12.8]). Let $\{W_n\}$ be a seq of independent r.v.s such that $\mathbb{E}(W_n) = 0$; $\sum \frac{\text{Var}(W_n)}{n^2} < \infty$

Then $\frac{1}{n} \sum_{k \leq n} W_k \rightarrow 0$ a.s.

Proof. Let $S_n = \sum_{k \leq n} W_k$ & $b_n = n$. then by Kronecker's Lemma 5.3.6, it is evident. □

Lemma 5.3.8 (Kolmogorov's Truncation Lemma [2, 12.9]). Suppose that $X_1, X_2, \dots$ i.i.d. r.v.s which have same distribution as X where $\mathbb{E}|X| < \infty$. Set $\mu := \mathbb{E}(X)$.

Define $Y_n = \begin{cases} X_n & \text{if } |X_n| \leq n \\ 0 & \text{if } |X_n| > n \end{cases}$. Then,

i) $\mathbb{E}(Y_n) \rightarrow \mu$

ii) $\mathbb{P}(Y_n = X_n \text{ eventually}) = 1$

iii) $\sum \frac{\text{Var}(Y_n)}{n^2} < \infty$

Proof. i) Define $Z_n = \begin{cases} X & \text{if } |X| \leq n \\ 0 & \text{if } |X| > n \end{cases}$ $Z_n \stackrel{D}{=} Y_n$. In particular $\mathbb{E}(Z_n) = \mathbb{E}(Y_n)$

As $n \rightarrow \infty$, $Z_n \rightarrow X$, $|Z_n| \leq |X|$ & $\mathbb{E}|X| < \infty$. Thus, by DCT,

$$\mathbb{E}(Z_n) \rightarrow \mu \implies \mathbb{E}(Y_n) \rightarrow \mu$$

ii) Now,

$$\begin{aligned} \sum_{n=1}^{\infty} \mathbb{P}(Y_n \neq X_n) &= \sum \mathbb{P}(|X_n| > n) = \sum \mathbb{P}(|X| > n) \\ &= \mathbb{E} \sum_{n=1}^{\infty} I_{(|X| > n)} = \mathbb{E} \left(\sum_{1 \leq n < |X|} 1 \right) \leq \mathbb{E}|X| < \infty \end{aligned}$$

By Borel Cantelli Lemma, $\mathbb{P}(Y_n = X_n \text{ eventually}) = 1$

iii)

$$\begin{aligned} \sum \frac{\text{Var}(Y_n)}{n^2} &\leq \sum \frac{\mathbb{Y}_{\infty}^{\neq}}{n^2} = \sum \frac{\mathbb{E}(|X|^2; |X| \leq n)}{n^2} = \mathbb{E}(|X|^2 f(|X|)) \\ \text{where } f(z) &= \sum \frac{1}{n^2} I_{(|Z| \leq n)} = \sum_{n \geq \max(1, z)} \frac{1}{n^2} \leq \frac{2}{\max(1, z)} \quad (\text{Using } \frac{1}{n^2} \leq \frac{2}{n(n+1)} = 2(\frac{1}{n} - \frac{1}{n+1})) \\ \therefore \sum \frac{\text{Var}(Y_n)}{n^2} &\leq 2\mathbb{E}|X| < \infty \end{aligned}$$

□

5.4 Doob's decomposition

Definition 5.4.1. We say that a process A is a previsible and null at 0 if A_n is $\mathcal{F}_{n-1}$ measurable for all n and $A_0 = 0$ respectively.

Theorem 5.4.2 ([2, 12.11]). a) $\{X_n : n \in \mathbb{Z}^+\}$ be an adapted process with $X_n \in \mathcal{L}^1 \forall n$. Then X has a Doob decomposition

$$X = X_0 + M + A \tag{5.4.1}$$

where M is a martingale null at 0 and A is a previsible process at 0.

Moreover, this decomposition is unique modulo indistinguishability in the sense that if $X = X_0 + \tilde{M} + \tilde{A}$ is another such decomposition then $\mathbb{P}(M_n = \tilde{M}_n, A_n = \tilde{A}_n \forall n) = 1$

b) X is a submartingale iff the process A is an increasing process in the sense that

$$\mathbb{P}(A_n \leq A_{n+1} \quad \forall n) = 1$$

Proof. If X has a Doob decomposition as at (5.4.1), then since M is a martingale & A is previsible, we have a.s.

$$\mathbb{E}(X_n - X_{n-1} | \mathcal{F}_{n-1}) = \mathbb{E}(M_n - M_{n-1} | \mathcal{F}_{n-1}) + \mathbb{E}(A_n - A_{n-1} | \mathcal{F}_{n-1}) = 0 + (A_n - A_{n-1})$$

Hence,

$$A_n = \sum_{k=1}^n \mathbb{E}(X_k - X_{k-1} | \mathcal{F}_{k-1}) \quad \text{a.s.} \tag{5.4.2}$$

and if we use (5.4.2) to define A , we obtain the required decomposition of X .

Let $X_0 + M + A = X_0 + \tilde{M} + \tilde{A} \implies Y_n = M_n - \tilde{M}_n = \tilde{A}_n - A_n$. Thus Y_n is martingale which is a previsible process. Hence $Y_n = 0$ a.s. As a result, the decomposition is unique.

b) Obvious from before.

□

5.5 The angle-brackets process $\langle M \rangle$

Lemma 5.5.1 ([2, 12.12]). *Let M be a martingale in $\mathcal{L}^2$ and null at 0.*

- a) M^2 is a submartingale.
- b) $M^2 = N + A$ where N is a martingale and A is a previsible increasing process, both N and A being null at 0 where we define $A_\infty = \uparrow \lim A_n$ a.s.
- c) M is bounded in $\mathcal{L}^2$ iff $\mathbb{E}A_\infty < \infty$

Proof. Using the conditional form of Jensen's inequality, we have a.s.

$$\begin{aligned} \mathbb{E}((M_n - M_{n-1})^2 | \mathcal{F}_{n-1}) &\geq (\mathbb{E}(M_n - M_{n-1}) | \mathcal{F}_{n-1})^2 \\ \implies \mathbb{E}(M_n^2 | \mathcal{F}_{n-1}) - M_{n-1}^2 &\geq 0 \\ \implies \mathbb{E}(M_n^2 | \mathcal{F}_{n-1}) &\geq M_{n-1}^2 \end{aligned}$$

Thus M^2 is a submartingale and has a Doob decomposition (essentially unique). This proves part (b). Using the martingale property of N , part (c) follow. $\square$

Remark 5.5.2. The process A is written $\langle M \rangle$. Note that $\mathbb{E}(M_n^2) = \mathbb{E}(A_n)$ and $A_n - A_{n-1} = \mathbb{E}(M_n^2 - M_{n-1}^2 | \mathcal{F}_{n-1}) = \mathbb{E}((M_n - M_{n-1})^2 | \mathcal{F}_{n-1})$.

5.6 Relating convergence of M to finiteness of $\langle M \rangle_\infty$

Let M be a martingale in $\mathcal{L}^2$ & null at 0. Define $A := \langle M \rangle$

Theorem 5.6.1 ([2, 12.13]). a) $\lim_n M_n(\omega)$ exists for a.e. ω for which $A_\infty(\omega) < \infty$

b) Suppose that M has uniformly bounded increments i.e.

$$|M_n(\omega) - M_{n-1}(\omega)| \leq K \quad \forall n, \omega$$

Then $A_\infty(\omega) < \infty$ for a.e. $\omega \implies \lim M_n(\omega)$ exists.

Proof. a) As A is previsible, it is immediate that $\forall k \in \mathbb{N}$,

$$S(k) := \inf\{n \in \mathbb{Z}^+ : A_{n+1} > k\} \quad \text{defines a stopping time.}$$

Stopped process $A^{S(k)}$ is previsible because for $B \in \mathcal{B}$ & $n \in \mathbb{N}$,

$$\begin{aligned} \{A_{n \wedge S(k)} \in B\} &= F_1 \cup F_2 \\ \text{where } F_1 &:= \cup_{r=0}^{n-1} \{S(k) = r; A_r \in B\} \in \mathcal{F}_{n-1} \\ F_2 &:= \{A_n \in B\} \cap \{S(k) \leq n-1\}^c \in \mathcal{F}_{n-1} \end{aligned}$$

Since $(M^{S(k)})^2 - A^{S(k)} = (M^2 - A)^{S(k)}$ is a martingale [Use of Uniqueness]. $\langle M^{S(k)} \rangle = A^{S(k)}$
Now the process $A^{S(k)}$ is bounded by K .

$$\therefore M^{S(k)} \text{ is bounded in } \mathcal{L}^2 \text{ and } \lim M_{n \wedge S(k)} \text{ exists a.s.} \quad (5.6.1)$$

However,

$$\{A_\infty < \infty\} = \cup_k \{S(k) = \infty\} \quad [S(k) = \infty = \inf(\phi) \implies A_{n+1} \leq K \forall n] \quad (5.6.2)$$

Now (5.6.1) + (5.6.2) implies that (a)

- b) Suppose that $\mathbb{P}(A_\infty = \infty, \sup_n |M_n| < \infty) > 0$
 Let $T(c) = \inf\{r : |M_r| > c\}$, then for some c ,

$$\mathbb{P}(A_\infty = \infty, T(c) = \infty) > 0 \quad (5.6.3)$$

Now, $\mathbb{E}(M_{T(c) \wedge n}^2 - A_{T(c) \wedge n})^2 = 0$ and $M^{T(c)}$ is bounded by $c + K$ [from given condition and stopping time]

$$\implies \mathbb{E}(A_{T(c) \wedge n}) \leq (c + K)^2 \quad \forall n \quad (5.6.4)$$

By MCT, (5.6.3) and (5.6.4) doesn't hold simultaneously. So (b) follows

By (5.6.4), $\mathbb{E}(A_{T(c) \wedge n}) \leq (c + K)^2$

Now,

$$\mathbb{E}(A_{T(c)}) \geq \mathbb{E}(A_{T(c)} I_B) = \mathbb{E}(A_\infty I_B) = \infty \cdot \mathbb{P}(B) = \infty \quad \text{where } B = \{A_\infty = \infty, T(c) = \infty\}$$

Thus this is a contradiction. □

5.7 ‘Strong law’ for martingales in $\mathcal{L}^2$

M is a martingale in $\mathcal{L}^2$ and null at 0. Let $A = \langle M \rangle$.

Since $(1 + A)^{-1}$ is bounded previsible process

$$W_n := \sum_{1 \leq k \leq n} \frac{M_k - M_{k-1}}{1 + A_k} = \left((1 + A)^{-1} \circ M \right)_n \quad \text{defines a martingale } W$$

Moreover, since $(1 + A_n)$ is $\mathcal{F}_{n-1}$ measurable ,

$$\begin{aligned} \mathbb{E}((W_n - W_{n-1})^2 | \mathcal{F}_{n-1}) &= \langle W \rangle_n - \langle W \rangle_{n-1} \\ &= (1 + A_n)^{-2} (A_n - A_{n-1}) \\ &\leq \frac{1}{1 + A_{n-1}} - \frac{1}{1 + A_n} \end{aligned}$$

We see that $\langle W \rangle_\infty \leq 1$ a.s. $\lim W_n$ exists a.s.

By Kronecker's lemma 5.3.6 , $\frac{M_n}{A_n} \rightarrow 0$ a.s on $\{A_\infty = \infty\}$

5.8 Levy's extension of Borel Cantelli Lemmas

Theorem 5.8.1 ([2, 12.15]). For $n \in \mathbb{N}, E_n \in \mathcal{F}_n$

Define

$$Z_n := \sum_{1 \leq k \leq n} I_{E_k} = \text{number of } E_k (k \leq n) \text{ which occur.}$$

Define $\xi_k := \mathbb{P}(E_k | \mathcal{F}_{k-1})$ &

$$Y_n := \sum_{1 \leq k \leq n} \xi_k$$

Then a.s.

$$a) (Y_\infty < \infty) \implies (Z_\infty < \infty)$$

$$b) (Y_\infty = \infty) \implies \left(\frac{Z_n}{Y_n} \rightarrow 1 \right)$$

Remark 5.8.2. i) $\mathbb{E}(\xi_k) = \mathbb{P}(E_k)$. If $\sum \mathbb{P}(E_k) < \infty$, then $Y_\infty < \infty$ a.s. Borel-Cantelli Lemma first part follows from (a).

- ii) $\{E_n : n \in \mathbb{N}\}$ is a seq of independent events associated with $(\Omega, \mathcal{F}, \mathbb{P})$
and define $\mathcal{F}_n := \sigma(E_1, \dots, E_n)$, then $\xi_k = \mathbb{P}(E_k)$ a.s. and Borel-Cantelli Lemma second part follows from (b).

Proof. Let M be the martingale $Z - Y$ so that $Z = M + Y$ is the Doob decomposition of the submartingale Z then,

$$\begin{aligned} A_n &:= \langle M \rangle_n = \sum_{k=1}^n \mathbb{E}[(M_k - M_{k-1})^2 | \mathcal{F}_{k-1}] \\ &= \sum_{k=1}^n \mathbb{E}[(I_{E_k} - \xi_k)^2 | \mathcal{F}_{k-1}] = \sum_{k=1}^n \mathbb{E}[I_{E_k} + \xi_k^2 - 2I_{E_k}\xi_k | \mathcal{F}_{k-1}] \\ &= \sum_{k=1}^n \xi_k(1 - \xi_k) \leq \sum_{k=1}^n \xi_k = Y_n \quad \text{a.s.} \end{aligned}$$

If $Y_\infty < \infty$ which implies $A_\infty < \infty$ and $\lim M_n$ exists so that Z_∞ is finite.

If $Y_\infty = \infty$ and $A_\infty < \infty$ then $\lim M_n$ exists and it is trivial that $\frac{Z_n}{Y_n} \rightarrow 1$

If $Y_\infty = \infty$ and $A_\infty = \infty$, $\frac{M_n}{A_n} \rightarrow 0$ so that $\frac{M_n}{Y_n} \rightarrow 0$ and $\frac{Z_n}{Y_n} \rightarrow 1$

□

Chapter 6

Uniform Integrability

6.1 Introduction

In this chapter, we discuss a property known as the uniform integrability and its implications.

The main reference for this chapter is [2, Chapter 13].

Lemma 6.1.1 ([2, 13.1]). *Suppose that $X \in \mathcal{L}^1 = \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. Then given $\epsilon > 0$, there exists a $\delta > 0$ such that for $F \in \mathcal{F}$, $\mathbb{P}(F) < \delta$ implies $\mathbb{E}(|X|; F) < \epsilon$*

Proof. Assuming otherwise, we know that for some $\epsilon_0 > 0$, we can find a seq. $\{F_n\} \in \mathcal{F}$ such that

$$\mathbb{P}(F_n) < 2^{-n} \quad \text{and} \quad \mathbb{E}(|X|; F_n) \geq \epsilon_0$$

Let $H := \limsup F_n$. Now, as $\sum \mathbb{P}(F_n) < \infty$, we get $\mathbb{P}(H) = 0$

Using Fatou's lemma, we get $\mathbb{E}(|X|; H) \geq \epsilon_0$ which is a contradiction. $\square$

Corollary 6.1.2 ([2]). *Suppose that $X \in \mathcal{L}^1$ and $\epsilon > 0$ there exists a $K \in [0, \infty)$ such that*

$$\mathbb{E}(|X|; |X| > K) < \epsilon$$

Proof. Let there exist a δ such that $\mathbb{P}(F) < \delta$ for $F \in \mathcal{F}$

Now $K\mathbb{P}(|X| > K) \leq \mathbb{E}(|X|)$ we choose K such that $\mathbb{P}(|X| > K) < \delta$ $\square$

6.2 UI Family

Definition 6.2.1 ([2]). A class $\mathcal{C}$ of r.v.s is called uniformly integrable (UI) if given $\epsilon > 0$, there exists a $K \in [0, \infty)$ such that

$$\mathbb{E}(|X|; |X| > K) < \epsilon \quad \forall X \in \mathcal{C}$$

We note that for such a class $\mathcal{C}$, we have (K_1 , relating to $\epsilon = 1$) for every $X \in \mathcal{C}$.

$$\mathbb{E}|X| = \mathbb{E}(|X|; |X| > K_1) + \mathbb{E}(|X|; |X| \leq K_1) \leq 1 + K_1$$

Thus a UI Family is bounded in $\mathcal{L}^1$. [Converse isn't true].

6.2.1 Simple sufficient conditions for the UI property

a) Suppose $\mathcal{C}$ is a class of r.v. which are bounded in $\mathcal{L}^p$ for some $p > 1$ and for some $A \in [0, \infty)$,

$$\mathbb{E}(|X|^p) < A \quad \forall X \in \mathcal{C}$$

then $\mathcal{C}$ is UI.

Proof. If $v \geq k > 0$, then $v \leq k^{1-p}v^p$.

Then for $k > 0$ and $X \in \mathcal{C}$, we have

$$\mathbb{E}(|X|; |X| > k) \leq k^{1-p} \mathbb{E}(|X|^p; |X| > k) \leq k^{1-p} A$$

Now, we take k^{1-p} to be as small as possible such that $k^{1-p}A = \epsilon$ □

b) Suppose $\mathcal{C}$ is a class of r.v.s which is dominated by an integrable non-negative Y i.e.

$$|X(\omega)| \leq Y(\omega) \quad \forall X \in \mathcal{C} \text{ and } \mathbb{E}(Y) < \infty$$

Then $\mathcal{C}$ is UI.

Proof. $\mathbb{E}(|X|; |X| > k) \leq \mathbb{E}(Y; |X| > k) \leq \mathbb{E}(Y; Y > k) < \epsilon$.

Now applying corollary 6.1.2 we get the following result. □

Theorem 6.2.2 ([2]). *Let $X \in \mathcal{L}^1$. Then the class $\{\mathbb{E}(X|\mathcal{G}) : \mathcal{G} \text{ is a sub } \sigma\text{-algebra of } \mathcal{F}\}$ is UI.*

Proof. Let $\epsilon > 0$. Choose $\delta > 0$ such that for $F \in \mathcal{F}$, $\mathbb{P}(F) < \delta$. Hence, $\mathbb{E}(|X|; F) < \epsilon$

$\mathcal{G}$ is a sub σ -algebra of $\mathcal{F}$ and Y be any version of $\mathbb{E}(X|\mathcal{G})$

By Jensen's equality,

$$|\mathbb{E}(X|\mathcal{G})| \leq \mathbb{E}(|X||\mathcal{G}) \text{ a.s.} \implies |Y| \leq \mathbb{E}(|X||\mathcal{G}) \text{ a.s.}$$

Hence, $\mathbb{E}|Y| \leq \mathbb{E}|X|$ a.s. Now using Markov inequality,

$$K\mathbb{P}(|Y| > K) \leq \mathbb{E}|Y| \leq \mathbb{E}|X| \implies \mathbb{P}(|Y| > K) < \frac{\mathbb{E}|X|}{K}$$

Now choose K such that $\frac{\mathbb{E}|X|}{K} < \delta$.

But $\{|Y| > K\} \in \mathcal{G} \subseteq \mathcal{F}$ so from Lemma 6.1.1 and definition of conditional expectation,

$$\mathbb{E}(|Y|; |Y| > K) \leq \mathbb{E}(|X|; |Y| > K) < \epsilon$$

□

Theorem 6.2.3 (Bounded Convergence Theorem). *[[2]] Let $\{X_n\}$ be a sequence of r.v.s and X be a r.v. Suppose $X_n \xrightarrow{P} X$ and for some $K \in [0, \infty)$, we have $|X_n(\omega)| \leq K$. Then $\mathbb{E}|X_n - X| \rightarrow 0$. Note that earlier we assumed a.s. convergence and now we assuming convergence in probability.*

Proof. Indeed for $t \in \mathbb{N}$,

$$\mathbb{P}(|X| > K + \frac{1}{t}) \leq \mathbb{P}(|X_n - X| > \frac{1}{t}) \quad \forall n$$

Thus, $\mathbb{P}(|X| > K + \frac{1}{t}) = 0$ Hence, $\mathbb{P}(|X| > K) = \mathbb{P}(\cup_t \{|X| > K + \frac{1}{t}\}) = 0$

For $\epsilon > 0$, $\mathbb{P}(|X_n - X| > \frac{\epsilon}{3}) < \frac{\epsilon}{3K}$ when $n \geq N_0$, we get

$$\begin{aligned} \mathbb{E}(|X_n - X|) &= \mathbb{E}(|X_n - X|; |X_n - X| > \frac{\epsilon}{3}) + \mathbb{E}(|X_n - X|; |X_n - X| \leq \frac{\epsilon}{3}) \\ &\leq 2K \frac{\epsilon}{3} + \frac{\epsilon}{3} \mathbb{P}(|X_n - X| \leq \frac{\epsilon}{3}) \leq \epsilon \end{aligned}$$

Thus, $\mathbb{E}(|X_n - X|) \rightarrow 0$ □

Remark 6.2.4. Applying Continuous Mapping theorem, we can get the result for $\mathcal{L}^p$

6.2.2 Necessary and sufficient condition for $\mathcal{L}^1$ convergence

Theorem 6.2.5 ([2]). *Let $\{X_n\}$ & $X \in \mathcal{L}^1$. Then $X_n \xrightarrow{\mathcal{L}^1} X$ equivalently, $\mathbb{E}(|X_n - X|) \rightarrow 0$ iff*

i) $X_n \xrightarrow{P} X$

ii) $\{X_n\}$ is UI.

Proof. If part: i) and ii) are satisfied.

For $K \in [0, \infty)$, define $\phi_K : \mathbb{R} \rightarrow [-K, K]$, $\phi_K(x) = \begin{cases} K & \text{if } x > K \\ x & \text{if } |x| \leq K \\ -K & \text{if } x < -K \end{cases}$ For $\epsilon > 0$. By the UI

property of $\{X_n\}$ and lemma 6.1.2,

$$\begin{aligned} & \mathbb{E}(|\phi_K(X_n) - X_n|) \\ &= \mathbb{E}(|\phi_K(X_n) - X_n|; |X_n| \leq K) + \mathbb{E}(|\phi_K(X_n) - X_n|; X_n > K) + \mathbb{E}(|\phi_K(X_n) - X_n|; X_n < -K) \\ &\leq 2K + \mathbb{E}(|X_n|; |X_n| > K) \end{aligned}$$

Choose K such that $\mathbb{E}(|\phi_K(X_n) - X_n|) < \frac{\epsilon}{3}$ and $\mathbb{E}(|\phi_K(X) - X|) < \frac{\epsilon}{3}$

But $|\phi_K(x) - \phi_K(y)| \leq |x - y|$. Therefore, $\phi_K(X_n) \xrightarrow{P} \phi_K(X)$

By theorem 6.2.3, there exists n_0 such that $\forall n \geq n_0$, $\mathbb{E}(|\phi_K(X_n) - \phi_K(X)|) < \frac{\epsilon}{3}$

Thus, by triangle equality, $\mathbb{E}(|X_n - X|) < \epsilon$ and the proof is complete.

Only If Part: $X_n \xrightarrow{P} X$ and $\{X_n\}$ is UI implies X_n is bounded in $\mathcal{L}^1$. Thus using the previous theorem 6.2.3, we get the result. $\square$

Chapter 7

UI Martingales

7.1 Introduction

Here we bring together the concepts of L^1 convergence and uniform integrability.

The main reference for this chapter is [2, Chapter 14].

Definition 7.1.1 ([2]). Let M be a UI martingale, so that M is a martingale relative to our setup $(\Omega, \mathcal{F}, \mathcal{F}_n, \mathbb{P})$ & $(M_n : n \in \mathbb{Z}^+)$ is a UI family.

Since M is UI, M is bounded in $\mathcal{L}^1$ and $M_\infty = \lim M_n$ exists a.s. using Doob's Forward Convergence Theorem. Now by (6.2.2), $M_n \rightarrow M_\infty$ in $\mathcal{L}^1$. Thus we are left to prove $M_n = \mathbb{E}(M_\infty | \mathcal{F}_n)$.

For $r \geq n$, the martingale property gives

$$\mathbb{E}(M_r | \mathcal{F}_n) = M_n \quad (7.1.1)$$

But for $F \in \mathbb{F}_n$, $|\mathbb{E}(M_r; F) - \mathbb{E}(M_\infty; F)| \leq \mathbb{E}(|M_r - M_\infty|; F) \leq \mathbb{E}(|M_r - M_\infty|)$ a.s.. Letting $r \rightarrow \infty$ in (7.1.1), we get, $\mathbb{E}(M_\infty | \mathcal{F}_n) = \mathbb{E}(M_n | \mathcal{F}_n) = M_n$

Theorem 7.1.2 ([2]). Let M be a UI martingale. Then $M_\infty = \lim M_n$ exists a.s. and in $\mathcal{L}^1$. Moreover, $\forall n, M_n = \mathbb{E}(M_\infty | \mathcal{F}_n)$ a.s.

Theorem 7.1.3 (Levy's 'Upward' Theorem). [[2]] Let $\xi \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and define $M_n := \mathbb{E}(\xi | \mathcal{F}_n)$. Then M is a UI martingale and $M_n \rightarrow \eta := \mathbb{E}(\xi | \mathcal{F}_\infty)$ a.s and in $\mathcal{L}^1$

Proof. We know that $\mathcal{F}_{n-1} \subseteq \mathcal{F}_n$, thus $\mathbb{E}(M_n | \mathcal{F}_{n-1}) = \mathbb{E}(\mathbb{E}(\xi | \mathcal{F}_n) | \mathcal{F}_{n-1}) = \mathbb{E}(\xi | \mathcal{F}_{n-1}) = M_{n-1}$. Hence M is a UI and $M_\infty := \lim M_n$ exists a.s. & in $\mathcal{L}^1$.

It is required to prove $M_\infty = \eta$ a.s. where $\eta := \mathbb{E}(\xi | \mathcal{F}_\infty)$. Without loss of generality, we assume that $\xi \geq 0$.

Consider 2 measures Q_1 and Q_2 on $(\Omega, \mathcal{F}_\infty)$ where

$$Q_1(F) := \mathbb{E}(\eta; F), Q_2(F) := \mathbb{E}(M_\infty; F) \quad , \quad F \in \mathcal{F}_\infty$$

If $F \in \mathcal{F}_n$, then since $\mathbb{E}(\eta | \mathcal{F}_\infty) = \mathbb{E}(\eta | \mathcal{F}_n)$, $\mathbb{E}(\eta; F) = \mathbb{E}(M_n; F) = \mathbb{E}(M_\infty; F)$. Thus Q_1 and Q_2 are on the π -system $\cup \mathcal{F}_n$ and hence they agree on $\mathcal{F}_\infty$.

Both η and M_∞ are $\mathcal{F}_\infty$ measurable. More strictly M_∞ may be taken to be $\mathcal{F}_\infty$ measurable by defining $M_\infty := \limsup M_n \quad \forall \omega$. Thus

$$F := \{\omega : \eta > M_\infty\} \in \mathcal{F}_\infty \text{ and since } Q_1(F) = Q_2(F)$$

Therefore, $\mathbb{E}(\eta - M_\infty; \eta > M_\infty) = 0$ Hence $\mathbb{P}(\eta > M_\infty) = 0$. Similarly, $\mathbb{P}(\eta < M_\infty) = 0$.

Thus, $\eta = M_\infty$ a.s. □

7.2 Martingale proof of Kolmogorov's 0-1 law

Theorem 7.2.1. *[[2]] $X_1, X_2, \dots$ are a sequence of independent r.v.s. Define $\tau_n := \sigma(X_{n+1}, X_{n+2}, \dots)$; $\tau := \bigcap_n \tau_n$.
Then if $F \in \tau$, $\mathbb{P}(F) = 0$ or 1*

Proof. Define $\mathcal{F}_n := \sigma(X_1, X_2, \dots, X_n)$. Let $F \in \tau$ and $\eta := I_F$. Since $\eta \in b\mathcal{F}_\infty$, Levy's upward theorem shows that $\eta = \mathbb{E}(\eta|\mathcal{F}_\infty) = \lim \mathbb{E}(\eta|\mathcal{F}_n)$ a.s.

$\forall n$, η is τ_n measurable and hence independent of $\mathcal{F}_n$. Therefore, $\mathbb{E}(\eta|\mathcal{F}_n) = \mathbb{E}(\eta) = \mathbb{P}(F)$ a.s. Hence $\eta = \mathbb{P}(F)$ a.s.

Now as η takes only 0 or 1, we have $\mathbb{P}(F) = 0$ or 1 □

Theorem 7.2.2 (Levy's 'Downward' Theorem). *[[2]] Suppose $(\Omega, \mathcal{F}, \mathbb{P})$ and that $\{\mathcal{G}_{-n} : n \in \mathbb{N}\}$ is a collection of sub σ -algebras of $\mathcal{F}$ s.t.*

$$\mathcal{G}_{-\infty} := \bigcap_k \mathcal{G}_{-k} \subseteq \dots \subseteq \mathcal{G}_{-(n+1)} \subseteq \mathcal{G}_{-n} \subseteq \dots \subseteq \mathcal{G}_{-1}$$

Let $\gamma \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and define $M_{-n} := \mathbb{E}(\gamma|\mathcal{G}_{-n})$. Then

$$M_{-\infty} := \lim M_{-n} \text{ exists a.s. and in } \mathcal{L}^1 \text{ and } M_{-\infty} = \mathbb{E}(\gamma|\mathcal{G}_{-\infty}) \text{ text a.s.} \quad (7.2.1)$$

Proof. Applying the Upcrossing Lemma 4.2.2 to the martingale

$$(M_k, \mathcal{G}_k : -N \leq k \leq -1)$$

we get, $\mathbb{E}(M_{-n}) = \mathbb{E}(\gamma) < \infty \quad \forall n$ i.e. $\sup \mathbb{E}(M_{-n}) < \infty$. Thus $M_{-\infty} := \lim M_{-n}$ exists a.s.

Using the theorem 6.2.2 that $\mathbb{E}(\gamma|\mathcal{G}_{-n})$ is a UI, M_{-n} is a UI. Hence, $M_{-\infty} := \lim M_{-n}$ exists in $\mathcal{L}^1$

Now, $M_{-\infty} := \limsup M_{-n} \in \mathcal{G}_{-\infty}$

Let $G \in \mathcal{G}_{-\infty} \in \mathcal{G}_{-r}$, $\mathbb{E}(\gamma; G) = \mathbb{E}(M_{-r}; G)$ and letting $r \uparrow \infty$, we get 7.2.1. □

7.3 Martingale proof of the Strong Law

Theorem 7.3.1 ([2]). *Let $X_1, X_2, \dots$ be i.i.d. r.v.s with $\mathbb{E}|X_k| < \infty \quad \forall k$; $\mu = \mathbb{E}(X_n) \quad \forall n$; $S_n := X_1 + \dots + X_n$*

Then $\frac{S_n}{n} \xrightarrow[\mathcal{L}^1]{a.s.} \mu$

Proof. $\mathcal{G}_{-n} := \sigma(S_n, S_{n+1}, \dots) := \sigma(S_n, X_{n+1}, X_{n+2}, \dots)$; $\mathcal{G}_{-\infty} := \bigcap_n \mathcal{G}_{-n}$

Now, $\sigma(X_{n+1}, X_{n+2}, \dots)$ is independent of $\sigma(X_1, S_n) \subseteq \sigma(X_1, \dots, X_n)$. Therefore, $\mathbb{E}(X_1|\mathcal{G}_{-n}) = \mathbb{E}(X_1|\mathcal{S}_n)$

Consider Λ to be the law of X , then $s_n = x_1 + \dots + x_n$, we have

$$\begin{aligned} \mathbb{E}(X_1; S_n \in B) &= \int \dots \int_{S_n \in B} x_1 \Lambda(dx_1) \Lambda(dx_2) \dots \Lambda(dx_n) \\ &= \mathbb{E}(X_2; S_n \in B) = \dots = \mathbb{E}(X_n; S_n \in B) \end{aligned}$$

Thus, $\mathbb{E}(X_1|S_n) = \dots = \mathbb{E}(X_n|S_n) = \frac{1}{n} \mathbb{E}(X_1 + \dots + X_n|S_n) = \frac{1}{n} S_n$ a.s. and hence, $\mathbb{E}(X_1|\mathcal{G}_{-n}) = \frac{1}{n} S_n$ a.s.

Now, $\mathbb{E}|X_k| < \infty$. Thus S_n is a martingale. Hence, $L = \lim \frac{S_n}{n}$ exists a.s. & in $\mathcal{L}^1$. For definiteness,

$L := \limsup \frac{S_n}{n} \quad \forall \omega$. For each k , $L = \limsup \frac{X_{k+1} + \dots + X_{k+n}}{n}$ so that $L \in m\tau_k$ where $\tau_k = \sigma(X_{k+1}, \dots)$

By Kolmogorov's 0-1 law 7.2.1, $\mathbb{P}(L = c) = 1$ for some c in $\mathbb{R}$. But $c = \mathbb{E}(L) = \lim \mathbb{E}(\frac{S_n}{n}) = \mu$ □

Theorem 7.3.2 (Doob's Submartingale Inequality). *[[2]] Let Z be a non-negative submartingale. Then for $c > 0$,*

$$c\mathbb{P}\left(\sup_{k \leq n} Z_k \geq c\right) \leq \mathbb{E}\left(Z_n; \sup_{k \leq n} Z_k \geq c\right) \leq \mathbb{E}(Z_n)$$

Proof. Let $F = \{\sup_{k \leq n} Z_k \geq c\}$; F is a disjoint union. $F = F_0 \cup F_1 \cup \dots \cup F_n$ where $F_0 := \{Z_0 \geq c\}$; $F_k := \{Z_0 < c\} \cap \{Z_1 < c\} \cap \dots \cap \{Z_k \geq c\}$. Now, $F_k \in \mathcal{F}_k$ and $Z \geq c$ on F_k . Hence, $\mathbb{E}(Z_n; F_k) \geq \mathbb{E}(Z_k; F_k) \geq c\mathbb{P}(F_k)$. Summing over k , we get

$$c\mathbb{P}(\sup_{k \leq n} Z_k \geq c) \leq \mathbb{E}(Z_n; \sup_{k \leq n} Z_k \geq c) \leq \mathbb{E}(Z_n)$$

□

Lemma 7.3.3 ([2]). *If M is a martingale, c is a convex function and $\mathbb{E}|c(M_n)| < \infty \forall n$, then $c(M)$ is a submartingale.*

Proof. Using the conditional form of Jensen's inequality, $c(M_{n-1}) = c(\mathbb{E}(M_n | \mathcal{F}_{n-1})) \leq \mathbb{E}(c(M_n) | \mathcal{F}_{n-1})$

□

Lemma 7.3.4 (Kolmogorov's Inequality). *[[2]] Let $(X_n : n \in \mathbb{N})$ be a sequence of independent zero-means r.v.s in $\mathcal{L}^2$. Define $\sigma_k^2 := \text{Var}(X_k)$; $S_n := X_1 + \dots + X_n$; $V_n := \text{Var}(S_n) = \sum_{k=1}^n \sigma_k^2$. Then for $c > 0$,*

$$c^2 \mathbb{P}(\sup_{k \leq n} |S_k| > c) \leq V_n$$

Proof. S_n is a martingale w.r.t $\mathcal{F}_n := \sigma(X_1, \dots, X_n)$

$\therefore S_n^2$ is a non-negative submartingale. Thus, using Doob's Submartingale inequality,

$$c^2 \mathbb{P}(\sup_{k \leq n} |S_k| > c) \leq \sum \mathbb{E}(S_n^2) = V_n$$

This completes the proof.

□

Lemma 7.3.5 (A consequence of Holder's Inequality). *[[2]] X and Y are non-negative r.v.s such that $c\mathbb{P}(x \geq c) \leq \mathbb{E}(Y; X \geq c) \forall c > 0$. Then for $p > 1$ and $\frac{1}{p} + \frac{1}{q} = 1$, we have $\|X\|_p \leq q\|Y\|_p$*

Theorem 7.3.6 (Doob's $\mathcal{L}^p$ inequality). *[[2]]*

- a) *Let $p > 1$ and define q so that $\frac{1}{p} + \frac{1}{q} = 1$. Let Z be a non-negative submartingale bounded in $\mathcal{L}^p$ and define $Z^* := \sup_{k \in \mathbb{Z}^+}$. Then, $Z^* \in \mathcal{L}^p$ and indeed,*

$$\|Z^*\|_p \leq \sup_r \|Z_r\|_p \tag{7.3.1}$$

The submartingale Z is therefore dominated by Z^ of $\mathcal{L}^p$. As $n \rightarrow \infty$; $Z_\infty := \lim Z_n$ exists a.s. and in $\mathcal{L}^p$ and $\|Z_\infty\|_p = \sup_r \|Z_r\|_p = \uparrow \lim_r \|Z_r\|_p$*

- b) *If Z is of the form $|M|$, where M is a martingale bounded in $\mathcal{L}^p$, then $M_\infty := \lim M_n$ exists a.s. and in $\mathcal{L}^p$ and of course $Z_\infty = |M_\infty|$*

Proof. For $n \in \mathbb{Z}^+$, define $Z_n^* := \sup_{k \leq n} Z_k$

Using Doob's Submartingale inequality, we get,

$$c\mathbb{P}(\sup_{k \leq n} Z_k \geq c) \leq \mathbb{E}(Z_n; \sup_{k \leq n} Z_k \geq c)$$

Thus, $c\mathbb{P}(Z_n^* \geq c) \leq \mathbb{E}(Z_n; Z_n^* \geq c)$

Using the previous lemma 7.3.5, $\|Z_n^*\|_p \leq q\|Z_n\|_p \leq q \sup_r \|Z_r\|_p$.

Now (7.3.1) follows after applying MCT.

Since $(-Z)$ is a supermartingale bounded in $\mathcal{L}^p$ implying that it is bounded in $\mathcal{L}^1$ too.

Hence, $Z_\infty = \lim Z_n$ exists a.s. (Using Doob's Forward Convergence theorem)

However, $|Z_n - Z|^p \leq (2Z^*)^p \in \mathcal{L}^p$ So DCT shows, $Z_n \rightarrow Z \in \mathcal{L}^p$

Jensen's inequality shows that $\|Z_r\|_p$ is non-decreasing in r and the rest is obvious.

□

Chapter 8

Application to Option Pricing

8.1 Introduction

In this chapter we discuss an application of discrete time martingale theory in option pricing. More specifically, we discuss the discrete version of the celebrated Black-Scholes formula.

The main reference for this chapter is [2, Chapter 15].

8.2 A martingale representation result

For $S = \{-1, 1\}$, let Σ be the set of all subsets of S . Fix $p \in (0, 1)$ and let μ be a probability measure on (S, Σ) with

$$\mu(\{1\}) = p = 1 - \mu(\{-1\})$$

Let $N \in \mathbb{N}$. Define $(\Omega, \mathcal{F}, \mathbb{P}) = (S, \Sigma, \mu)^N$. A typical element of Ω is $\omega = (\omega_1, \dots, \omega_N)$, $\omega_k \in \{-1, 1\}$. Define $\varepsilon_k : \Omega \rightarrow \mathbb{R}$ by $\varepsilon_k(\omega) = \omega_k$ so that $(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_N)$ are i.i.d. r.v.s with law μ .

For $0 \leq n \leq N$, define

$$Z_n := \sum_{k=1}^n (\varepsilon_k - 2p + 1) ; \mathcal{F}_n := \sigma(Z_1, \dots, Z_n) = \sigma(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n)$$

Note that $\mathbb{E}(\varepsilon_k) = 1 \cdot p + (-1)(1-p) = 2p-1$. Here, Z is a martingale relative to $(\{\mathcal{F}_n : 0 \leq n \leq N\}, \mathbb{P})$

Lemma 8.2.1 ([2]). *If M is a martingale relative to $(\{\mathcal{F}_n\}, \mathbb{P})$, then $\exists$ a unique previsible process H s.t. $M = M_0 + H \circ Z$ i.e. $M_n = M_0 + \sum_{k=1}^n H_k(Z_k - Z_{k-1})$*

Remark 8.2.2. $M_0 = \mathbb{E}(M_n)$ and is constant on Ω as $\mathcal{F}_0 = \{\phi, \Omega\}$

Proof. We construct H explicitly. M_n is $\mathcal{F}_n$ measurable.

Therefore, $M_n(\omega) = f_n(\varepsilon_1(\omega), \varepsilon_2(\omega), \dots, \varepsilon_n(\omega)) = f_n(\omega_1, \dots, \omega_n)$ for some function $f_n : \{-1, 1\}^n \rightarrow \mathbb{R}$. Since M is a martingale,

$$\begin{aligned} 0 &= \mathbb{E}(M_n - M_{n-1} | \mathcal{F}_n)(\omega) = pf_n(\omega_1, \dots, \omega_{n-1}, 1) + (1-p)f_n(\omega_1, \dots, \omega_{n-1}, -1) + f_{n-1}(\omega_1, \dots, \omega_{n-1}) \\ \implies &\frac{f_n(\omega_1, \dots, \omega_{n-1}, 1) - f_{n-1}(\omega_1, \dots, \omega_{n-1})}{2(1-p)} = \frac{f_{n-1}(\omega_1, \dots, \omega_{n-1}) - f_n(\omega_1, \dots, \omega_{n-1}, -1)}{2p} \end{aligned}$$

Define $H_n(\omega)$ to be their common value. H is previsible.

We are to now check the uniqueness of H . Suppose, there exists a H' which fulfils the same conditions.

$$\forall n \quad M_n = M_0 + \sum_{k=1}^n H_k(Z_k - Z_{k-1}) = M_0 + \sum_{k=1}^n H'_k(Z_k - Z_{k-1})$$

Now $\forall n \quad H_n = H'_n$ as Z_i 's are non-trivial. Thus H is unique. □

8.3 Option pricing ; discrete Black-Scholes formula

Consider there are two ‘securities’: *bonds* of fixed interest rate r and *stock*, the value of which fluctuates randomly. We assume that the value of these changes only at times $1, 2, \dots, N$ where $N \in \mathcal{N}$. For $n = 0(1)N$, we write

1. $B_n = (1 + r)^n B_0$ for the value of 1 bond unit throughout $(n, n + 1)$
2. S_n for the value of 1 unit of stock throughout $(n, n + 1)$

At time 0, if one starts with fortune x invested in A_0 units of stock and V_0 units of bond, so that

$$A_0 S_0 + V_0 B_0 = x$$

Between time 0 and 1, after rearranging the portfolio, we have A_1 units of stock and V_1 units of bond, so that

$$A_1 S_0 + V_1 B_0 = x$$

So at time $n - 1$, we have

$$X_{n-1} = A_{n-1} S_{n-1} + V_{n-1} B_{n-1}$$

Thus, just before time n after the rearrangement of the portfolio, the fortune is described by

$$X_{n-1} = A_n S_{n-1} + V_n B_{n-1} \quad (n \geq 1).$$

The fortune just after n is given by

$$X_n = A_n S_n + V_n B_n \quad (n \geq 0) \tag{8.3.1}$$

and the change in fortune is

$$X_n - X_{n-1} = A_n (S_n - S_{n-1}) + V_n (B_n - B_{n-1}) \tag{8.3.2}$$

Now, $B_n - B_{n-1} = r B_{n-1}$ and $S_n - S_{n-1} = R_n S_{n-1}$, where R_n is the random ‘rate of interest of stock at time n ’. We rewrite (8.3.2) as

$$X_n - X_{n-1} = r X_{n-1} + A_n S_{n-1} (R_n - r) \tag{8.3.3}$$

so that on setting $Y_n = (1 + r)^{-n} X_n$ we get

$$Y_n - Y_{n-1} = (1 + r)^{-(n-1)} A_n S_{n-1} (R_n - r) \tag{8.3.4}$$

Note that Y_n is the discounted value of the fortune at time n , so the increase in Y_n is of our primary interest.

Let $\Omega, \mathcal{F}, \varepsilon_n$ ($1 \leq n \leq N$), Z_n ($0 \leq n \leq N$) and $\mathcal{F}_n$ ($0 \leq n \leq N$) be as in Section 8.1.

We build a model in which each R_n takes only values a, b in $(-1, \infty)$, where $a < r < b$ by setting

$$R_n = \frac{a + b}{2} + \frac{b - a}{2} \varepsilon_n$$

Then $R_n - r = \frac{1}{2}(b - a)(\varepsilon_n - 2p + 1) = \frac{1}{2}(b - a)(Z_n - Z_{n-1})$ where $p := \frac{r-a}{b-a}$

A *European option* is a contract made just after time 0 which will allow you to buy 1 unit of stock just after time N at a price K , also known as striking price. If you have made such a contract, then just after time N , you will exercise the option if $S_N > K$ and will not if $S_N < K$. Thus, the value at time N of such a contract is $(S_N - K)^+$.

A hedging strategy with initial value x for the described option is a portfolio management scheme $\{(A_n, V_n) : 1 \leq n \leq N\}$ where the processes A and V are previsible with respect to $\{\mathcal{F}_n\}$ and where, with X satisfying (8.3.1) and (8.3.2), we have for every ω ,

1. $X_0(\omega) = x$,
2. $X_n(\omega) \geq 0 \quad (0 \leq n \leq N)$,
3. $X_n(\omega) = (S_N(\omega) - K)^+$

Theorem 8.3.1 ([2]). *A hedging strategy with initial value x exists if and only if*

$$x = x_0 := \mathbb{E}[(1+r)^{-N}(S_N - K)^+]$$

where the expectation is taken with respect to the measure $\mathbb{P}$ as described in Section 8.1 with p as taken above. There is a unique hedging strategy with initial value x_0 , and it involves no short-selling, i.e. A is never negative.

Proof. Suppose that a hedging strategy with initial value x exists and let A, V, X, Y denote the associated processes. Clearly, $Y = Y_0 + F \circ Z$ where F is the previsible process with

$$F_n = (1+r)^{-(n-1)} A_n S_{n-1}$$

Of course, F is bounded. Thus Y is a martingale under the $\mathcal{P}$ measure, since Z is; and since $Y_0 = x$ and $Y_N = (1+r)^{-N}(S_N - K)^+$, we get $x = x_0$ by (8.3.3) and definition of hedging strategy. Let us define

$$Y_n := \mathbb{E}[(1+r)^{-N}(S_N - K)^+ | \mathcal{F}_n]$$

then Y is a martingale and using the result from Section 8.2, we notice that for some unique previsible process A , (8.3.4) holds. Thus defining $X_n := (1+r)^n Y_n$ and $V_n := \frac{X_n - A_n S_n}{B_n}$, (8.3.1) and (8.3.2) holds. Since

$$X_0 = x \text{ and } X_N = (S_N - K)^+$$

we need to just prove that A is never negative. If we use the structure of H in the Section 8.2, we have

$$\mathbb{E}[(S_N - K)^+ | S_{n-1}, S_n = (1+b)S_{n-1}] \geq \mathbb{E}[(S_N - K)^+ | S_{n-1}, S_n = (1+a)S_{n-1}].$$

The result follows. □

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